

Separating News from Entertainment: Attention and Political Accountability

Marta Korczak¹

¹Utrecht University

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Abstract

How effective is public scrutiny when exposure to political information becomes elective? I model media technology as the price of steering attention away from a media platform's default mixture. Lowering that price makes entertainment-leaning voters consume less news, and news-leaning voters more. Because these responses mirror each other, aggregate scrutiny depends on the skew of tastes for content: balanced electorates are unaffected, however dispersed, while scrutiny falls where average interest in news is low, and most where tastes are homogeneous. Unbundling benefits every voter privately but can reduce welfare, since nobody internalizes the fall in public good.¹

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1 Introduction

When news and entertainment are consumed together, exposure to political information is partly incidental. Back in the days of TV with just a few channels, viewers had to either wait to watch a movie broadcast after the daily news or watch it with a movie. In the digital era, they can easily consume movies via streaming platforms. Therefore, the cost of adjusting media consumption to the individually optimal has fallen, and the exposure to less preferred content has become elective. How can these developments have an impact on political accountability?

A large empirical literature shows that media environments shape voter information and political accountability. Greater newspaper circulation increases government responsiveness (Besley and Burgess 2002), while reduced media coverage weakens electoral discipline (Snyder and Strömberg 2010). Digital technologies can similarly alter political participation and policy outcomes (Gavazza et al. (2019)), and often substitute for traditional sources while improving consumer welfare when access remains free (Gentzkow 2007). While these findings establish that media technologies influence political behavior, less is understood about how the ability to separate news from entertainment shapes aggregate monitoring and political incentives. In this paper, I show that scrutiny can worsen once exposure to news is a matter of choice and voters prefer to consume entertainment. Accountability then depends less on the availability of political information than on the composition of the demand for it. As such, this paper formalizes empirical findings of Prior (2005).

I study a two-period electoral accountability model with heterogeneous voters, a strategic incumbent, and free entry by media platforms. Each voter divides attention between news and entertainment, and differs in her intrinsic taste for news relative to entertainment. Every media firm supplies some default mixture of news and entertainment, and a voter who wants a diet different from that default exerts a *steering cost*, proportional to the informational distance between his preferred diet and the default. A bundled environment — a newspaper, a broadcast bulletin — has a high steering cost: editorial choices are made on the reader's behalf, and departing from them is relatively expensive. An unbundled environment — a streaming platform, a news website — has a low steering cost: the reader assembles his own mix almost freely.

Attention to the news produces scrutiny. Aggregate scrutiny disciplines the incumbent, who exerts costly first-period effort, and selects between types at the election. The resulting public good is non-rival: each voter enjoys it undivided but internalizes only a small fraction of her own contribution to it. On the supply side, news is produced whenever attention is larger than a critical mass determined by fixed newsroom costs, and in that case, every voter consumes his preferred amount of news.

The first result characterizes the *technology gap*: the change in a voter's news consumption when the environment moves from bundled to unbundled. There is no gap for voters with no interest in news, for voters interested in nothing else, and for voters with tastes who consume the default mix under either technology. In other cases, unbundling reduces the news consumption of every entertainment-leaning voter and raises that of every news-leaning one, whatever the default. The technology gap is the largest not at the extremes but at intermediate tastes.

In the case when for every voter disliking news there is a voter liking in the same proportion — meaning, electorates are balanced — there is no change in aggregate scrutiny at all: the news abandoned by one wing is exactly offset by the news taken up by the other. Aggregate scrutiny falls if and only if the average interest in news falls short of the midpoint. Dispersion matters only in interaction with that skew, and it matters in a specific direction: among electorates with the same average taste, the one that loses most scrutiny is the one in which every voter *has* that taste. Strongly divided media diets, therefore, mute the consequences of unbundling, while uniform, mildly disengaged electorates lose most of the scrutiny.

On the incumbent’s side, the two functions of elections come apart. Greater scrutiny always raises effort: more monitoring always buys more first-period public good. But the same response makes types harder to tell apart, because bad incumbents with low effort costs imitate good ones, and the share doing so grows linearly in scrutiny. The informational advantage of a good incumbent is therefore single-peaked, maximized at an intermediate level of attention — a “Goldilocks” region — and eroded when attention becomes very high. Monitoring strengthens incentives and weakens selection at the same time.

When news is separated from entertainment, every voter is strictly better off in private terms: a lower steering price lets each obtain a content diet closer to his own taste, and only voters with exactly balanced tastes are left indifferent. While each voter has a limited impact on the public good, he enjoys it undivided — scrutiny is non-rival, yet his own contribution to it is diluted at a rate $1/N$, where N is the number of voters. He therefore internalizes the whole of the private gain and a vanishing share of the collective loss, so the individually optimal response to a cheaper technology is collectively excessive. Unbundling consequently faces no private opposition: no voter would choose to keep the bundle unless it happens to be his favourite content mix. Aggregate welfare can nonetheless fall, especially in the case of a homogeneous electorate that, on average, prefers entertainment.

2 Related literature

This paper contributes to the theoretical literature on the political economy of media by examining how the lower cost of obtaining a bundle of news and entertainment shapes political accountability. A large body of work studies how voter attention—often modeled through rational inattention—affects political behavior and policy outcomes. This literature shows that limits to attention influence the implementation of reforms (Prato and Wolton 2018; Hu and Li 2019; Devdariani and Hirsch 2023; Blumenthal 2023; Blumenthal 2025), pandering incentives (Trombetta 2020), political polarization (Hu et al. 2023; Matějka and Tabellini 2021), and electoral outcomes (Martinelli 2006; Bruns and Himmler 2016). Comprehensive reviews are provided by Maćkowiak et al. (2023) and Balles et al. (2025).

Most rational inattention models conceptualize political information as costly to acquire. Here information is free, and what is costly is departing from whatever mix the medium supplies by default. The relevant primitive is therefore not the price of becoming informed, but the price of *choosing*, and the object of interest is not the level of attention but the composition of the citizens who retain it. This has the following consequences: a technology change that is Pareto-improving in private terms while potentially collectively harmful, an aggregate effect that is invariant to the dispersion of tastes and responds only to their skew,

and a set of pivotal voters located at intermediate positions. The model’s demand system is observationally equivalent to a CES specification, but the implied share parameter would have to move with the technology — which is an argument against taking CES preferences as the primitive, and for taking the cost of steering as the primitive instead.

The paper also relates to the broader literature on political agency and electoral discipline. In the canonical framework of [Ferejohn \(1986\)](#), elections mitigate moral hazard by allowing voters to sanction poorly performing incumbents. Subsequent work highlights the role of the media in improving political outcomes by facilitating information transmission and reducing capture ([Besley and Prat 2006](#)). In contrast to models where media content is shaped primarily by ideological incentives or competitive pressures ([Gentzkow and Shapiro 2006](#); [Mullainathan and Shleifer 2005](#)), the media environment studied here links news provision directly to voters’ demand for attention.

Finally, the analysis connects to research examining how media markets interact with political outcomes, particularly in settings where informational specialization is limited. [Perego and Yuksel \(2022\)](#) show that smaller communities may exhibit lower polarization, suggesting that ideological differentiation plays a reduced role in such environments. By abstracting from ideological bias, this paper isolates the attention channel through which media technology affects electoral incentives. More broadly, the results contribute to a growing literature emphasizing heterogeneity in political agency by demonstrating that the distribution of voters’ preferences governs the effectiveness of electoral monitoring.

3 Model setting

The game lasts two periods. There are N voters indexed by $J = 1, \dots, N$. Voters are heterogeneous with respect to the taste parameter for news α_J drawn from a beta distribution: $\alpha_J \sim \text{Beta}(a, b) \forall J \in N$. The taste parameter for entertainment is equal to $1 - \alpha_J$. Therefore, in this setting, I assume voters consume only these two types of content. I interpret news as any content that provides information about an incumbent, and entertainment as everything else. In the baseline specification, voters are the same in all dimensions but the taste parameter α . Additionally, the model includes media producers and a ruler. The latter can be of two types: good or bad, with exogenous probability of being good γ . If she is good, her cost of putting in effort is zero. Otherwise, her cost is drawn from a uniform distribution: $c \sim U[0, 1]$. The ruler decides how much effort to put into both periods, and the more effort she puts in, the more likely she is to be re-elected. The political environment features both selection and incentive problems, but the central distortion studied here arises from moral hazard, as incumbent effort is unobserved.

3.1 Timing

In the first period, a ruler (she) is randomly selected. She can be either of a good type ($\theta = g$) with probability γ , or of a bad type ($\theta = b$). Only the ruler knows about her type. Each consumer J (he) draws his taste parameter α_J and chooses the amount of entertainment $t_{J,e}^*$ and news $t_{J,n}^*$ to consume, with an “attention budget” constraint: $t_{J,n}^* + t_{J,e}^* = 1$. The average demands for entertainment and news are given by $t_e^* = \frac{\sum_{J=1}^N t_{J,e}^*}{N}$.

and $t_n^* = \frac{\sum_{j=1}^N t_{j,n}^*}{N}$, respectively. Subsequently, M media producers decide to enter the market. Their number depends on the demand for media content, on exogenously determined advertising revenues, and on fixed costs for entertainment and news. Each media producer operates under the same profit function, and they cannot target specific consumers (i.e., their offer is homogenous). Their choice is discrete: whether to fund a newsroom (i.e., produce news on top of entertainment).

Following entry and production decisions, voters consume their preferred amount of entertainment and news (if a newsroom is founded). News is supplied exclusively by media firms and produced at the maximum-demand level $t_n^s = \max_J t_{J,n}$ (therefore, no consumer is rationed in news). Note that $1 - \tilde{t}_{J,n}$ denotes total realized entertainment consumption by consumer J , including content obtained from outside the modeled media sector: the consumer is also not rationed in entertainment. The term $\min\{1 - t_{J,n}^*, 1 - t_n^s\}$ represents in-model entertainment consumption, with any residual interpreted as an outside option. If the newsroom is founded, the average realized news consumption is $\tilde{t}_n = \frac{1}{N} \sum_J \tilde{t}_{J,n}$, which always equals average desired news consumption, t_n^* .

Subsequently, if a ruler is of a bad type, she chooses at once the optimal amount of effort she puts in the first e_1 and second e_2 period, with $e_p \in [0, 1]$.² The public good in period p is proportional to the incumbent's effort: $e_p \tau_p$. It is non-rival, and enters into consumers' utility function. The maximum amount of the public good is always larger in the second period ($\tau_1 < \tau_2$). Public-good provision depends on political tenure rather than calendar time. Parameter τ_1 denotes the maximum public good produced by a first-term politician, while $\tau_2 > \tau_1$ applies only to an incumbent who is re-elected and therefore serves a second term. A newly elected replacement, even if entering in the second calendar period, is a first-term officeholder and thus generates public good τ_1 . The political environment features tenure-based learning or capacity, so that experienced incumbents can generate a higher public good ($\tau_2 > \tau_1$).

An incumbent serves a first term, knowing how much attention consumers pay to news and entertainment. Then, consumers consume media content. The more they pay attention to the news, the more motivated the incumbent is to exert effort. The probability of re-election is given by $\rho(\tilde{t}_n, e_1) = \tilde{t}_n \sqrt{e_1} + (1 - \tilde{t}_n) \rho_0$ where ρ_0 is probability of re-election if a voter is not informed. This reduced form captures the idea that higher average news consumption improves voter monitoring and therefore increases the probability that poor-performing incumbents are replaced, while having a smaller effect on high-performing incumbents. Appendix A shows the microfoundation underlying this specification.

Finally, elections are held during which consumers decide whether to re-elect the incumbent. If they do, the second period of the game begins, in which the politician puts effort e_2 into producing the public good. Note that consumers only obtain a noisy signal about the incumbent's performance; the revelation of the realized amount of public good, $\tau_1 e_1 + \tau_2 e_2$, happens after the game is concluded.

²There is no difference if the ruler chooses the effort consecutively, as after the second period there are no re-election incentives. A good incumbent will choose maximum effort in the second period, and a bad incumbent will choose zero effort, regardless of the timing of the decision.

3.2 Consumers/Voters

There are N voters indexed by $J = 1, \dots, N$, each allocating a unit attention budget between news, $t_{J,n} \in [0, 1]$, and entertainment, $1 - t_{J,n}$. Voters are heterogeneous with respect to the taste parameter for news $\alpha_J \sim \text{Beta}(a, b)$, with population mean $\mu = \frac{a}{a+b}$ (the average interest in news). A media diet $t_{n,J}$ delivers content utility:

$$V(t_{n,J}; \alpha_J) = t_{n,J} \ln \alpha_J + (1 - t_{n,J}) \ln(1 - \alpha_J) = \ln(1 - \alpha_J) + x_J t_{n,J}, \quad (1)$$

where $x_J \equiv \ln \frac{\alpha_J}{1-\alpha_J}$ is the taste log-odds.

Media technology: a default bundle and a price of steering. A media environment is summarized by two objects. The first is its *default bundle* $\beta \in (0, 1)$: the share of news a viewer receives if he exerts no effort to steer — the composition of the broadcast schedule, or of the feed she is served. The second is the *price of steering* $\kappa > 0$: obtaining a diet other than the default requires seeking out content, resisting what is served, and switching, and κ prices that effort. Formally, the cost of a diet $t_{n,J}$ is its deviation from the default:

$$C_{\kappa,\beta}(t) = \kappa \left[t \ln \frac{t}{\beta} + (1 - t) \ln \frac{1-t}{1-\beta} \right] \geq 0, \quad (2)$$

which vanishes only at $t = \beta$. Going with the flow (the default bundle) is free; departing from the default is costly in proportion to how far the diet departs from it. The steering cost (2) shares its functional form with the costs used in the rational-inattention literature, but applies it to a different friction.³

The public good. Apart from content value net steering costs, a consumer gets utility from the non-rival public good produced by an incumbent (which I interpret as *governance quality*). Incumbent's effort e_p in period p delivers a public good $e_p \tau_p$, with $\tau_2 > \tau_1$, so that an experienced officeholder is more productive than a first-term one. Denote $g(\tilde{t}_n)$ for its expected total over the two periods, taken over the incumbent's type and cost draw. Aggregate scrutiny raises governance quality through two channels: it disciplines a bad incumbent into exerting effort, and it raises the probability that a good one is retained. Two features of this public good shape the voter's problem.

First, it is nonrival. Every voter enjoys the level delivered, undivided and irrespective of her own media diet: a citizen who watches only entertainment benefits from competent government exactly as much as one who follows every story. Second, while the benefit is undivided, a voter's influence over it is not. Her own attention moves aggregate scrutiny only at a rate $1/N$, so the effect of her diet on $g(\tilde{t}_n)$ is of order $1/N$. In other words, benefits are shared in full while influence is diluted, and monitoring of a politician is a by-product of media tastes, not an objective of theirs. Formally, the expected public good over two periods

³The specification here is close to the costly-control specification of [Mattsson and Weibull \(2002\)](#), in which an agent pays a cost to implement an action that differs from the status quo. Here, the media environment supplies both the default and the cost of changing it. Relatedly, [Matějka and Tabellini \(2021\)](#) study an agent whose cost is reducing uncertainty about a given policy, measured by entropy.

is:

$$\begin{aligned}
 g(\tilde{t}_n) = & \gamma \left[\underbrace{\rho_g(\tilde{t}_n, e_1^* = 1) \times (\tau_1 + \tau_2)}_{\text{good, re-elected: works both periods}} + \underbrace{(1 - \rho_g(\tilde{t}_n, e_1^* = 1)) \times (\tau_1 + \gamma\tau_1)}_{\text{good, replaced: works in } t=1, \text{ challenger delivers } \gamma\tau_1} \right] \\
 & + (1 - \gamma) \mathbb{E}_c \left[\underbrace{\rho_b(\tilde{t}_n, e_1^*) \times \tau_1 e_1^*}_{\text{bad, re-elected: shirks in } t=2} + \underbrace{(1 - \rho_b(\tilde{t}_n, e_1^*)) \times (\tau_1 e_1^* + \gamma\tau_1)}_{\text{bad, replaced: challenger delivers } \gamma\tau_1} \right], \tag{3}
 \end{aligned}$$

which simplifies to

$$g(\tilde{t}_n) = \gamma \left[\tau_1 + \rho_g(\tilde{t}_n, e_1^* = 1) \times \tau_2 + (1 - \rho_g(\tilde{t}_n, e_1^* = 1)) \times \gamma\tau_1 \right] + (1 - \gamma) \left[\tau_1 \mathbb{E}_c[e_1^*] + (1 - \mathbb{E}_c[\rho_b]) \gamma\tau_1 \right]. \tag{4}$$

where γ is the exogenous probability of an incumbent being of a good type ($\theta = g$), $\rho_\theta(\tilde{t}_n, e_1^*)$ is the endogenous probability of re-election of an incumbent of type θ , and c is the cost of exerting an effort by a bad incumbent. Appendix B studies the extension in which voters differ in their own valuation of the public good $g(\tilde{t}_n)$.

The voter's payoff in full. Voter J chooses her news share $t_{J,n} \in (0, 1)$ to maximize

$$W_J = \underbrace{t_{J,n} \ln \alpha_J + (1 - t_{J,n}) \ln(1 - \alpha_J)}_{\text{content value}} - \underbrace{\kappa \left[t_{J,n} \ln \frac{t_{J,n}}{\beta} + (1 - t_{J,n}) \ln \frac{1 - t_{J,n}}{1 - \beta} \right]}_{\text{steering cost } C_{\kappa, \beta}(t_{J,n})} + \underbrace{g(\tilde{t}_n)}_{\text{public good}}, \tag{5}$$

s.t.

$$t_{n,J} \in [0, 1], \quad \alpha_J \sim \text{Beta}(a, b), \quad \kappa > 0, \quad \beta \in (0, 1), \quad g'(\tilde{t}_n) > 0, \quad g''(\tilde{t}_n) < 0. \tag{6}$$

3.3 Media firms

Media firms sell attention to advertisers. Let $A_n > A_e > 0$ denote advertising revenue per unit of attention devoted to news and to entertainment. The premium reflects exclusivity: local reporting is produced by the modelled firms alone, whereas entertainment is a commodity, replicable and available from many sources, which is why it commands the lower rate. Attention itself is inelastic in total: a voter allocates a unit budget between the two, so where news is not reported the whole budget goes to entertainment.⁴ Firms operate platforms that carry both content types and monetize whatever attention they attract. The bundling of content is captured on the demand side, by the steering cost C_κ of Section 3.2. I do not impose any further format restriction on the supply side, which would represent the same friction twice. A firm's content decision is thus a single discrete one: whether to fund a newsroom, at fixed cost FC_n , in addition to the fixed cost FC_e of operating the platform itself. This distinction between the two costs is that entertainment can be licensed or replicated cheaply, whereas reporting requires a newsroom whose cost does not scale down with the size of the audience it serves. Equivalently, the newsroom decision is a restricted choice of default mix: a platform with a newsroom serves the default β , one without serves $\beta = 0$. Discreteness, therefore, reflects the assumption that firms choose between two defaults rather than over a continuum.

Denote $\nu = \{0, 1\}$ as an indicator whether the newsroom is funded. With $M \in \mathbb{N}$ symmetric

⁴If the modelled platforms capture only a share $\sigma \leq 1$ of entertainment attention, the incremental return becomes $(A_n - \sigma A_e)t_n^*$ and the critical mass $T = M FC_n / [(A_n - \sigma A_e)N]$. Since $\sigma \leq 1$, the specification in the text is the conservative case, with the highest threshold for viability.

platforms sharing the market, per-firm profit given the newsroom decision ν is

$$\Pi(\nu; t_n^*, M) = \frac{N}{M} \left[A_n \nu t_n^* + A_e (1 - \nu t_n^*) \right] - FC_e - \nu FC_n, \quad (7)$$

or equivalently

$$\Pi(\nu; t_n^*, M) = \frac{N}{M} \left[A_e + \nu (A_n - A_e) t_n^* \right] - FC_e - \nu FC_n. \quad (8)$$

Entertainment revenue A_e is gained on all attention regardless of ν . Funding a newsroom gives the premium $A_n - A_e$ on the share of attention devoted to the news, t_n^* .

Viability and entry. A firm without a newsroom earns A_e per unit of attention. Funding one converts each unit its audience devotes to news into revenue A_n , so the incremental return is the premium $A_n - A_e$ on aggregate news attention. With M symmetric firms sharing the market, news is produced if and only if

$$\mathbb{E}_F[t^*(\alpha; \kappa; \beta)] \geq T, \quad T \equiv \frac{M FC_n}{(A_n - A_e) N}, \quad (9)$$

where T is the *critical mass* of news demand required to sustain reporting. Free entry determines M as the largest integer at which per-firm profit is nonnegative.

3.4 Incumbent

I assume that a ruler has risk-neutral preferences and knows how much news voters consume. In each period in office, she receives a remuneration of r . If an elected politician is good, she faces no costs of exerting effort, $c = 0$. If she is of a bad type, her cost is drawn from a uniform distribution $c \sim U(0, 1)$ (before solving the optimization problem). Each type chooses an amount of effort to put in both periods:

$$\begin{aligned} e_1^*, e_2^* &= \arg \max_{e_1, e_2} \{ (1 - c(\theta) e_1) r + (1 - c(\theta) e_2) r \rho(t_n^*, e_1) \} \\ &\quad \text{s.t.} \\ e_1, e_2 &\in [0, 1]; \quad r > 0 \\ \rho(t_n^*, e_1) &= t_n^* \sqrt{e_1} + \rho_0(t_n^*, e_1) \\ c(\theta = g) &= 0; \quad c(\theta = b) \sim U(0, 1) \end{aligned} \quad (10)$$

The game can equivalently be formulated as sequential. For the bad incumbent, since second-period effort does not affect re-election prospects, it yields no strategic benefit, implying $e_2^* = 0$. The bad incumbent, therefore, effectively chooses only first-period effort.

3.5 Equilibrium Concept

I focus on pure strategies, and the equilibrium is a Perfect Bayesian Equilibrium. On an equilibrium path, the ruler chooses the level of effort, taking the attention of consumers to news as given, and consumers divide their attention between news and entertainment, correctly foreseeing the effort of an incumbent, given $Pr(\theta)$ and the distribution of c . Because payoffs depend only on realized consumption, I analyze equilibrium using a reduced game in realized choices. The appendix includes the definition of equilibrium along with proofs of its existence (Appendix C).

4 Solution

In equilibrium, three mechanisms determine political accountability, understood as the probability of re-electing a good incumbent and ousting a bad one.

- (i) **Incumbent effort.** A good type exerts full effort in both periods, while a bad type exerts positive effort only in the first, with intensity increasing in average attention to news $\tilde{t}_n$ and decreasing in her cost of effort.
- (ii) **Probability of re-election.** An incumbent is re-elected with probability $\rho(\tilde{t}_n, e_1) = \tilde{t}_n \sqrt{e_1} + (1 - \tilde{t}_n)\rho_0$, a mixture of what an informed electorate would deliver — which responds to effort — and what an uninformed one would, which does not. Higher news consumption strengthens the link between conduct and reelection.
- (iii) **Consumer demand and media supply.** Voters divide their attention according to their intrinsic taste for news α_J , the price of steering κ , and the default bundle β ; each also internalizes, at rate $1/N$, the effect of her own attention on the public good. Media firms fund a newsroom, and thereby supply the news that voters demand, whenever overall demand meets the critical mass required to cover its fixed cost.

While voters' attention directly disciplines a bad incumbent, discipline is not the same as selection: effort exerted to avoid removal also makes a bad type resemble a good one. Accountability is therefore stronger when the news is more attractive (larger $\mathbb{E}[\alpha]$, *ceteris paribus*) only where the incumbent is good or where a bad incumbent exerts little effort. “Too much” interest in news need not be optimal if it induces a bad incumbent to exert enough effort to be retained.

Equilibrium characterization. For every $\kappa > 0$, $\beta \in (0, 1)$ and $\alpha_J \sim \text{Beta}(a, b) \forall J \in N$, the game of Section 3 has a unique perfect Bayesian equilibrium, with the following properties.

- (i) **Voters.** Each voter's demand for news $\tilde{t}_{J,n}$ is interior and satisfies

$$t_{J,n}^*(\alpha_J; \kappa, \beta, \tilde{t}_n) = \frac{\alpha_J^{1/\kappa} \beta e^\zeta}{\alpha_J^{1/\kappa} \beta e^\zeta + (1 - \alpha_J)^{1/\kappa} (1 - \beta)} \in (0, 1), \quad \zeta \equiv \frac{g'(\tilde{t}_n)}{N\kappa}, \quad (11)$$

increasing in her taste α_J , in the default β , and in the civic term ζ , which measures the marginal effect of her own attention on the public good. Since ζ is of order $1/N$, this term is small in a large electorate, and (19) reduces to a function of $(\alpha_J, \kappa, \beta)$ alone.

- (ii) **Media firms.** Free entry determines the number of platforms M ; a newsroom is funded if and only if demand meets the critical mass, $\mathbb{E}_F[t_n^*] \geq T$ with $T = M FC_n / [(A_n - A_e)N]$. Where news is produced, no voter is rationed, so realized average scrutiny solves the fixed point

$$\tilde{t}_n = \begin{cases} \mathbb{E}_F[t_n^*(\alpha; \kappa, \beta, \tilde{t}_n)] & \text{if } \mathbb{E}_F[t_n^*] \geq T, \\ 0 & \text{otherwise.} \end{cases} \quad (12)$$

- (iii) **Incumbent.** A good type faces no cost of effort and exerts $e_1 = e_2 = 1$; a bad type with cost draw $c \sim U(0, 1)$ exerts

$$e_1^*(c; \tilde{t}_n) = \min \left\{ 1, \left(\frac{\tilde{t}_n}{2c} \right)^2 \right\}, \quad e_2^* = 0,$$

second-period effort carrying no electoral return. Integrating over c , expected first-period effort is $\mathbb{E}_c[e_1^*] = \tilde{t}_n - \tilde{t}_n^2/4$.

- (iv) **Election.** The incumbent is retained with probability $\rho(\tilde{t}_n, e_1) = \tilde{t}_n \sqrt{e_1} + (1 - \tilde{t}_n)\rho_0$, so that $\rho_g = \tilde{t}_n + (1 - \tilde{t}_n)\rho_0$ for a good type and $\mathbb{E}_c[\rho_b] = \frac{\tilde{t}_n^2}{2}(1 - \ln \frac{\tilde{t}_n}{2}) + (1 - \tilde{t}_n)\rho_0$ for a bad one.
- (v) **Consistency.** Voters' expectations are correct: the marginal public good $g'(\tilde{t}_n)$ entering (19) is evaluated at the incumbent's equilibrium effort from (iii).

A more formal definition of equilibrium, together with existence and uniqueness, is given in Appendix C; uniqueness requires limited feedback from aggregate scrutiny to individual diets, which holds whenever $N\kappa$ is not too small. Here I provide a step-by-step solution. Because voters' diets depend on aggregate scrutiny only through $g'(\tilde{t}_n)$, and the incumbent's effort depends on $\tilde{t}_n$ alone, I solve the incumbent's problem first, taking realized average news as given, and then close the model with (12).

4.1 Incumbent's turn.

As a good incumbent has zero cost of exerting an effort, and she wants to be re-elected, she maximizes her effort in period one. In period two, she is indifferent between any value of effort (I assume she exerts again the largest possible effort, $e_2 = 1$). An optimal effort of a bad type in period two is zero (there is no incentive to exert any effort, as it is not possible to be re-elected after the second period). In period one, the optimal effort is given by:

$$\begin{aligned} \frac{\partial \rho(\tilde{t}_n, e_1)}{\partial e_1} &= \frac{c}{1 - ce_2} \\ \text{If } e_2 = 0 \text{ and } \rho(\tilde{t}_n, e_1) &= \tilde{t}_n \sqrt{e_1} + (1 - \tilde{t}_n)\rho_0, \text{ we have:} \\ e_1^* &= \left(\frac{\tilde{t}_n}{2c} \right)^2 \end{aligned} \tag{13}$$

A bad incumbent's effort thus increases with the average amount of news, but decreases faster with the cost of effort. Her welfare, if re-elected, is $(1 - ce_1^*)r + r$ if she is of a bad type, and $2r$ if she is good. If not re-elected, the bad type gets $(1 - ce_1^*)r$, and the good type gets r . In particular, the absence of a corner at zero effort follows from the bounded support of the cost distribution.

4.2 Media firms' turn

The newsroom decision. A decision to produce news is discrete, so it is settled by a comparison. From (8), the incremental profit from funding a newsroom is

$$\Pi(1; t_n^*, M) - \Pi(0; t_n^*, M) = \underbrace{\frac{N}{M} (A_n - A_e) t_n^*}_{\text{advertising premium on news attention}} - \underbrace{FC_n}_{\text{cost of reporting}}. \tag{14}$$

Funding a newsroom does not create attention — voters have a unit budget either way — but it converts the share t_n^* of that attention from the entertainment rate A_e to the news rate A_n . The firm therefore reports news if and only if the premium earned on the audience it would attract covers the newsroom, that is, if and only if aggregate news demand meets a critical mass, as defined in (9).

Proposition 1 (News-market collapse). *Write $D_B = \mathbb{E}_F[t^*(\alpha; \kappa_B, \beta)]$ and $D_U = \mathbb{E}_F[t^*(\alpha; \kappa^U, \beta)]$ for average news demand under the bundled and unbundled environments, with population preferring entertainment over news ($\mathbb{E}_F[\alpha] < 0.5$). The technology gap is therefore defined as $\Delta_{gap} = D_B - D_U$. Let T denote the critical mass required to sustain a newsroom, as in (9). Suppose news is viable under bundling, $D_B \geq T$. Then:*

- (i) *News production ceases after unbundling if and only if $T \in (D_U, D_B]$.*
- (ii) *This set is nonempty if and only if $\Delta_{gap} > 0$, and it is an interval of critical-mass values of width exactly Δ_{gap} .*

Proof. (i) By the viability condition (9), news is produced in an environment with demand D if and only if $D \geq T$. Production therefore occurs under bundling and ceases under unbundling if and only if $D_B \geq T$ and $D_U < T$ hold jointly, i.e. $D_U < T \leq D_B$.

(ii) The interval $(D_U, D_B]$ is nonempty precisely when $D_U < D_B$, i.e. $\Delta_{gap} > 0$: the population must prefer entertainment, so that unbundling reduces rather than raises news demand. Its length is $D_B - D_U = \Delta_{gap}$.

□

If voters prefer entertainment, the unbundling reduces scrutiny by the marginal amount Δ_{gap} ; but for those whose critical mass lies in the interval above, the same shift in tastes removes news provision altogether. Oftentimes, at most one outlet is viable in a sufficiently small market (e.g., a town), the relevant threshold there is $FC_n/[(A_n - A_e)N]$, which is decreasing in N : small communities require higher per-capita news demand to sustain a newsroom, and are therefore the first to lose one if demand decreases or the ratio of advertisement revenues to costs.

Entry. Per-firm profit is strictly decreasing in the number of platforms, since each serves a share $1/M$ of a fixed audience while paying its fixed costs in full. Free entry therefore determines M as the largest integer at which profit remains non-negative,

$$M = \left\lfloor \frac{N[A_e + \nu(A_n - A_e)\bar{t}]}{FC_e + \nu FC_n} \right\rfloor,$$

and if even a single platform cannot break even, $M = 0$, and no media market operates.

Realized scrutiny. Because news carries no marginal cost once the newsroom exists, a firm that reports makes news freely available, and no voter is rationed. Realized average scrutiny is therefore

$$\tilde{t}_n = \begin{cases} t_n^*, & \text{if } t_n^* \geq T, \\ 0, & \text{otherwise,} \end{cases} \quad (15)$$

where t_n^* is the demand determined in Section 4.3. Media firms are thus demand-followers: they neither shape the composition of what voters consume nor ration it, and their only influence on political accountability runs through the discrete decision of whether reporting exists at all.

4.3 Voters' turn

First-order condition. Voter J chooses his news share to maximize content value net of steering cost, internalizing the effect of his attention on the public good. Differentiating (1), the marginal content value of news is his taste log-odds, $V'(t_{J,n}) = \ln \alpha_J - \ln(1 - \alpha_J) = x_J$. Differentiating the steering cost (2) term by term, the constants cancel and

$$C'_{\kappa,\beta}(t_{J,n}) = \kappa \left[\ln \frac{t_{J,n}}{1-t_{J,n}} - \bar{b} \right], \quad C''_{\kappa,\beta}(t_{J,n}) = \frac{\kappa}{t_{J,n}(1-t_{J,n})} > 0, \quad \bar{b} \equiv \ln \frac{\beta}{1-\beta}. \quad (16)$$

The objective is therefore strictly concave, and the marginal steering cost diverges to $\mp\infty$ at the corners, so the maximizer is interior for every $\alpha_J \in (0, 1)$. The first-order condition equates the two marginal benefits of news with its marginal cost:

$$\underbrace{x_J}_{\text{content value}} + \underbrace{\frac{g'(\tilde{t}_n)}{N}}_{\text{own contribution to the public good}} = \underbrace{\kappa \left[\ln \frac{t_{J,n}}{1-t_{J,n}} - \bar{b} \right]}_{\text{marginal cost of steering}}. \quad (17)$$

A voter steers his attention toward the news until the taste gains from one more unit, together with the marginal scrutiny it produces, equals the effort of obtaining it.

The media diet. Rearranging (17) gives the model's main relation:

$$\ln \frac{t_{J,n}^*}{1-t_{J,n}^*} = \underbrace{\frac{x_J}{\kappa}}_{\text{own taste}} + \underbrace{\bar{b}}_{\text{default}} + \underbrace{\zeta(\tilde{t}_n)}_{\text{civic term}}, \quad \zeta \equiv \frac{g'(\tilde{t}_n)}{N\kappa} \quad (18)$$

The log-odds of a voter's diet decompose additively into three terms: his own taste, scaled by the price of steering; where the environment points; and the value of the scrutiny his attention produces. Only the first varies across voters — the default and the civic term are common — and the third is of order $1/N$, so that in a large electorate diets are mostly governed by tastes and the environment alone. Solving (18),

$$t_{J,n}^*(\alpha_J; \kappa, \beta, \tilde{t}_n) = \frac{\alpha_J^{1/\kappa} \beta e^\zeta}{\alpha_J^{1/\kappa} \beta e^\zeta + (1 - \alpha_J)^{1/\kappa} (1 - \beta)}. \quad (19)$$

The civic motive is observationally equivalent to facing a slightly more news-oriented default β : internalizing the public good and being served more news produce the same diets.

Pass-through of tastes into diets. Differentiating (18) with respect to the taste log-odds,

$$\frac{\partial t_{J,n}^*}{\partial x_J} = \frac{t_{J,n}^*(1 - t_{J,n}^*)}{\kappa}, \quad (20)$$

which measures how strongly a difference in taste translates into a difference in what voters actually consume. Two voters with distinct tastes end up with diets that differ in proportion to (20): where it is small, people who disagree about content nonetheless watch much the same things; where it is large, the same disagreement produces sharply different media diets.

First, pass-through is inversely proportional to the price of steering. Where steering is expensive, even sharply differing tastes produce similar diets, because nobody finds it worth the effort to depart far from what is served; where steering is cheap, the same tastes produce widely differing diets. The media technology therefore sets the rate at which heterogeneity in *preferences* is converted into heterogeneity in *consumption* — and it is this conversion rate, not preferences themselves, that the digital transition changed.

Second, pass-through is uneven across voters: the factor $t_{J,n}^*(1 - t_{J,n}^*)$ is largest for voters whose diets are close to balanced and vanishes toward the corners, so the environment moves least those already strongly preferring to one content type.

Lemma 2 (Dual CES representation, inspired by [Anderson et al. \(1992\)](#)). *Fix $\kappa > 0$, $\beta \in (0, 1)$ and ζ . Demand (19) coincides with that of a voter maximizing the CES objective $((1 - a)(1 - t)^q + at^q)^{1/q}$ over $t \in (0, 1)$, with substitution parameter $q = 1 - \kappa$ and share parameter a satisfying*

$$\frac{a}{1 - a} = \frac{\alpha_J}{1 - \alpha_J} \times \left(\frac{\beta e^\zeta}{1 - \beta} \right)^\kappa. \quad (21)$$

Proof. The CES first-order condition is $at^{q-1} = (1 - a)(1 - t)^{q-1}$, i.e. $\frac{t}{1-t} = \left(\frac{a}{1-a} \right)^{1/(1-q)}$; setting $\kappa = 1 - q$ and substituting (21) reproduces (18). Appendix D gives the steps in full. $\square$

The lemma locates the model relative to the specification in which media technology is an elasticity of substitution in preferences: for instance, the bundled benchmark $\kappa = 2$ corresponds to CES complements ($q = -1$), the unbundled benchmark $\kappa = \frac{1}{2}$ to CES substitutes ($q = \frac{1}{2}$), and the two generate identical behavior. It also identifies the difficulty with that specification. Equation (21) holds the environment fixed; except in the symmetric case $\beta e^\zeta = \frac{1}{2}$, the share parameter representing a given voter depends on the technology. The same CES preference, therefore, cannot describe him in both environments, so a change in technology would have to be recorded as a change in his taste for news. The present formulation avoids this: tastes are the weights α_J , fixed across regimes, which are described by parameters (κ, β) ,

5 Effects of unbundling of news and entertainment

5.1 Technology gap

This section discusses the *technology gap*, briefly discussed in Proposition 1, which I define as the change in voters' attention to news when the environment shifts from bundled (interpreted as traditional media) to unbundled (digital media). This subsection analyses individual technology gaps, subsection 5.2 describes aggregation across voters and its consequences to overall scrutiny.

Lemma 3 (Shape of the technology gap). *Let $\kappa_B > \kappa_U > 0$ and $\beta \in (0, 1)$. Diets are evaluated in the large-electorate limit, where the civic term $\zeta = g'(\tilde{t}_n)/(N\kappa)$ vanishes, so that $t_n^*(\alpha; \kappa, \beta)$ depends on the environment only through (κ, β) . Define the technology gap*

$$h(\alpha) \equiv t_n^*(\alpha; \kappa_U, \beta) - t_n^*(\alpha; \kappa_B, \beta).$$

Then:

- (i) $h(0) = h(\frac{1}{2}) = h(1) = 0$, and these are the only zeros. The gap vanishes in the extreme cases — no interest ($\alpha = 0$) or exclusive interest ($\alpha = 1$) — and for perfectly balanced tastes ($\alpha = \frac{1}{2}$).
- (ii) $h < 0$ on $(0, \frac{1}{2})$ and $h > 0$ on $(\frac{1}{2}, 1)$: unbundling reduces the news consumption of voters who lean towards entertainment and raises that of voters who lean towards news, whatever the default bundle β .
- (iii) Write $h(\alpha; \beta)$ to display the dependence on the default. For every $\alpha, \beta \in (0, 1)$,

$$h(1 - \alpha; 1 - \beta) = -h(\alpha; \beta).$$

Interchanging the roles of news and entertainment — in the voter's taste and in the default bundle — reverses the technology gap.

- (iv) Suppose the bundled technology compresses tastes while the unbundled one amplifies them, $\kappa_B \geq 1 \geq \kappa_U$ with at least one inequality strict. If the default mix is symmetric, $\beta = \frac{1}{2}$, then h is strictly convex on $(0, \frac{1}{2})$ with a unique interior minimizer $\alpha^\dagger$, and strictly concave on $(\frac{1}{2}, 1)$ with a unique maximizer at $1 - \alpha^\dagger$.

Proof in Appendix E.

Remark 4 (The civic term at finite N). Not taking into account the civic term in Lemma 3 (i) does have consequences. Since $\zeta = g'(\tilde{t}_n)/(N\kappa)$ depends on κ directly, the two environments (with finite N and very large $N \rightarrow \infty$) differ by $\zeta_U/\zeta_B = (\kappa_B/\kappa_U)(g'(\tilde{t}_n^U)/g'(\tilde{t}_n^B))$. Cheaper steering makes the impact of the civic term stronger, but the difference is relatively small: at $N = 60$ and $g' = 2$ one obtains $h(\frac{1}{2}) = +0.012$, with the zero at 0.492 rather than 0.500 and the minimum at $\alpha^\dagger = 0.195$ rather than 0.200; the depth $h(\alpha^\dagger) = -0.2745$ is unchanged to five decimals. Both displacements are $O(1/N)$ and vanish at $N = 200$ already ($h(\frac{1}{2}) = +0.004$). The zeros at $\alpha = 0$ and $\alpha = 1$ hold for every N .

Taste parameter α determines the magnitude of change in consumed news after transitioning from traditional (bundled) to digital (unbundled) media. On the left side in Figure 1, taking the symmetric default $\beta = \frac{1}{2}$, the bundled response curve lies above the 45-degree line on $(0, \frac{1}{2})$ and is concave there, while the unbundled curve lies below it and is convex. On the right, their difference $h(\alpha) = t_U^* - t_B^*$ is therefore convex on $(0, \frac{1}{2})$, negative below the midpoint and positive above it, with a unique minimizer at $\alpha^\dagger \approx 0.20$ where a voter changes about 0.275 of her attention budget from news to entertainment.

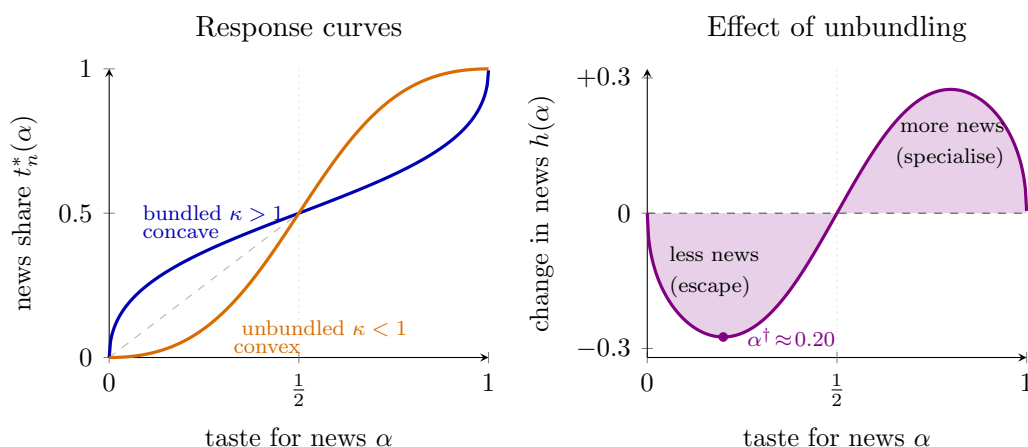


Figure 1: **Effects of unbundling, symmetric default** ($\beta = \frac{1}{2}$)

Remark 5 (Extreme technologies). Assuming $N \rightarrow \infty$, as $\kappa \rightarrow \infty$, steering is prohibitively costly and every diet is essentially at the default, $t_n^* \rightarrow \beta$, regardless of taste. As $\kappa \rightarrow 0$ steering is free, and since content value is linear in the diet each voter consumes only his preferred content $t_n^* \rightarrow \mathbf{1}\{\alpha > \frac{1}{2}\}$. The default becomes irrelevant in this limit. The effect of moving between these extremes is therefore $h_\infty(\alpha) = \mathbf{1}\{\alpha > \frac{1}{2}\} - \beta$: unbundling removes from an entertainment-leaning voter exactly the incidental news the bundle supplied, β , and grants a news-leaning voter exactly the news it withheld, $1 - \beta$. Aggregating, $\Delta_\infty(F) = 1 - p - \beta$ with $p = \Pr(\alpha < \frac{1}{2})$, so scrutiny becomes this extreme case precisely when a share $p > 1 - \beta$ of the electorate prefers entertainment.

Remark 5 describes the aggregate effect in the extreme cases of steering costs κ , and in the next subsection, I describe these effects for any $\kappa_B > \kappa_U$.

5.2 Aggregation: the role of the distribution of preferences

Assume throughout this section that the two environments, bundled and unbundled media satisfy $\kappa_B > \kappa_U$. Tastes are distributed $\alpha \sim \text{Beta}(\vartheta\mu, \vartheta(1-\mu))$, with mean μ and concentration $\vartheta > 0$; larger ϑ means a more homogeneous electorate. The technology gap h describes how one voter responds to unbundling, but what matters for accountability is the response of the electorate. Define the aggregate effect:

$$\Delta(F) \equiv \int_0^1 h(\alpha) f(\alpha) d\alpha = \mathbb{E}_F[t_n^*(\alpha; \kappa_U, \beta)] - \mathbb{E}_F[t_n^*(\alpha; \kappa_B, \beta)], \quad (22)$$

as the change in average news consumption when the environment moves from bundled to unbundled, holding tastes fixed. Since the N tastes in a model are i.i.d. draws from F , (22) is the expected change for every N ; the large-electorate limit is used only later, to evaluate g at the average diet rather than in expectation. Because h is negative below $\frac{1}{2}$ and positive above, $\Delta(F)$ nets a loss of scrutiny among entertainment-leaning voters against a gain among news-leaning ones, and its sign is not obvious a priori.

It becomes obvious once the electorate is read in mirror-image pairs. Consider two voters with tastes α and $1 - \alpha$: one leans towards entertainment by exactly as much as the other leans towards news. If the default mix is symmetric ($\beta = \frac{1}{2}$), Lemma 3 (iii) says unbundling treats them as exact opposites, $h(1 - \alpha) = -h(\alpha)$: the first loses precisely as much news as the second gains, so the pair contributes nothing to the aggregate. Were the two sides of every pair equally populated, all such contributions would cancel and $\Delta(F)$ would vanish however large the individual responses. Substituting α with $1 - \alpha$ on $(\frac{1}{2}, 1)$ makes the cancellation explicit:

$$\Delta(F) = \int_0^{1/2} h(\alpha) [f(\alpha) - f(1 - \alpha)] d\alpha. \quad (23)$$

Expressions (22) and (23) are the same number, but the second highlights that what matters for the change in scrutiny is the imbalance in its population weight, $f(\alpha) - f(1 - \alpha)$. The aggregate, therefore, depends on the taste distribution through its asymmetry around the default mix β , and the next proposition specifies what this implies.

Proposition 6 (Distributional effects of unbundling). *Let $\kappa_B > \kappa_U > 0$ and $\beta \in (0, 1)$. Parts (i)–(ii) hold for every default mix β ; part (iii) is stated for every $\alpha \sim \text{Beta}(\vartheta\mu, \vartheta(1 - \mu))$; part (iv) requires $\beta \leq \frac{1}{2}$.*

(i) **One-sided electorates.** If no voter leans towards news — F supported on $(0, \frac{1}{2}]$ and not degenerate at $\frac{1}{2}$ — then $\Delta(F) < 0$. Symmetrically, if no voter leans towards entertainment, $\Delta(F) > 0$. The default mix β plays no role.

(ii) **Balanced electorates and the default.** If the CDF of voters' tastes F is symmetric around $\frac{1}{2}$, then

$$\text{sign } \Delta(F) = \text{sign } \left(\frac{1}{2} - \beta\right),$$

whatever the dispersion of F . Neutrality (zero technology gap), therefore, requires two symmetries, of tastes and of the default: $\Delta(F) = 0$ for every symmetric F if and only if $\beta = \frac{1}{2}$. An entertainment-leaning default ($\beta < \frac{1}{2}$) leaves a balanced electorate consuming more news after unbundling, because bundling was holding it below its preferred mix. Moreover $\Delta(\tilde{F}; 1 - \beta) = -\Delta(F; \beta)$, so it suffices to study $\beta \leq \frac{1}{2}$.

(iii) **Threshold.** For $\alpha \sim \text{Beta}(\vartheta\mu, \vartheta(1 - \mu))$, Δ vanishes as $\mu \rightarrow 0$ and as $\mu \rightarrow 1$ and crosses zero exactly once in between: there is a unique $\mu^*(\beta, \vartheta)$ with $\Delta < 0$ for $\mu < \mu^*$ and $\Delta > 0$ for $\mu > \mu^*$. The threshold is increasing in β , satisfies $\mu^*(\frac{1}{2}, \vartheta) = \frac{1}{2}$, and converges to $\frac{1}{2}$ as $\vartheta \rightarrow \infty$ for every β . Unbundling lowers aggregate scrutiny if and only if average interest in news falls short of μ^* .

(iv) **Homogeneity deepens the loss.** Let $\beta \leq \frac{1}{2}$, fix $\mu < \frac{1}{2}$, and suppose tastes lie in $(0, \frac{1}{2}]$. Since $\Delta = \mathbb{E}_F[h(\alpha)]$ and h is strictly convex there,

$$|\Delta| \leq |h(\mu)|,$$

with equality only if every voter has taste exactly μ , and any mean-preserving spread strictly reduces $|\Delta|$.

Proof in Appendix F.

Corollary 7 (General taste distributions). Let $\beta = \frac{1}{2}$ and suppose F is entertainment-skewed: $f(\alpha) \geq f(1 - \alpha)$ for all $\alpha \in (0, \frac{1}{2})$, and $F(\frac{1}{2}) > \frac{1}{2}$, so that a strict majority of voters lean towards entertainment. Define the net entertainment-leaning share and the imbalance distribution

$$\pi = 2F(\frac{1}{2}) - 1, \quad \tilde{F}(\alpha) = \frac{1}{\pi} \int_0^\alpha [f(s) - f(1 - s)] ds \quad \text{on } (0, \frac{1}{2}), \quad \tilde{\mu} = \mathbb{E}_{\tilde{F}}[\alpha],$$

Then

$$\Delta = \pi \mathbb{E}_{\tilde{F}}[h(\alpha)] \quad \text{and} \quad |\Delta| \leq \pi |h(\tilde{\mu})|.$$

No restriction on the support of F is required. Proposition 6(iv) is the special case $\pi = 1$.

Proof of Corollary 7. By hypothesis $\pi = 2F(\frac{1}{2}) - 1 > 0$, and $f - \check{f} \geq 0$ on $(0, \frac{1}{2})$, so $\tilde{f}$ is non-negative and integrates to one there: it is a probability density.

At $\beta = \frac{1}{2}$ the gap is antisymmetric, $h(1 - \alpha) = -h(\alpha)$ (Lemma 3(iii)), so the general fold (54) becomes

$$\Delta = \int_0^{1/2} [h(\alpha)f(\alpha) + h(1 - \alpha)f(1 - \alpha)] d\alpha = \int_0^{1/2} h(\alpha)[f(\alpha) - f(1 - \alpha)] d\alpha = \pi \int_0^{1/2} h(\alpha) \tilde{f}(\alpha) d\alpha,$$

which is $\Delta = \pi \mathbb{E}_{\tilde{f}}[h(\alpha)]$.

By Lemma 3(iv), h is strictly convex on $(0, \frac{1}{2})$, so Jensen's inequality gives $\mathbb{E}_{\tilde{f}}[h] \geq h(\tilde{\mu})$, strictly unless $\tilde{f}$ is degenerate. Multiplying by $\pi > 0$ and using $h < 0$ there (Lemma 3(ii)) converts this to $|\Delta| \leq \pi |h(\tilde{\mu})|$.

Finally, if F is supported on $(0, \frac{1}{2}]$ then $f(1 - \alpha) = 0$ for $\alpha \in (0, \frac{1}{2})$, so $\pi = 1$, $\tilde{f} = f$ and $\tilde{\mu} = \mu$, and the bound reduces to Proposition 6(iv). $\square$

Remark 8 (The Beta family). The family $\alpha \sim \text{Beta}(\vartheta\mu, \vartheta(1 - \mu))$ assigns positive probability to tastes above $\frac{1}{2}$ however small μ is, so Proposition 6(iv) does not apply to it directly. Corollary 7 does apply, and its two hypotheses can be checked directly. First, the density ratio is

$$\frac{f(\alpha)}{f(1 - \alpha)} = \left(\frac{\alpha}{1 - \alpha}\right)^{\vartheta(2\mu - 1)}.$$

For $\alpha \in (0, \frac{1}{2})$ the base $\alpha/(1 - \alpha)$ is below one, and $\mu < \frac{1}{2}$ makes the exponent negative, so the ratio exceeds one: of any two voters with mirror-image tastes, the entertainment-leaning one is the more numerous. Second, substituting α with $1 - \alpha$ gives $1 - F(\frac{1}{2}) = \int_0^{1/2} f(1 - \alpha) d\alpha$, so

$$2F(\frac{1}{2}) - 1 = \int_0^{1/2} [f(\alpha) - f(1 - \alpha)] d\alpha > 0,$$

and a strict majority of the electorate lies below the midpoint. Both conditions of the corollary hold, so $|\Delta| \leq \pi |h(\tilde{\mu})|$. The two extremes of concentration limit the range. As $\vartheta \rightarrow \infty$ voters become identical, $\pi \rightarrow 1$ and $\tilde{\mu} \rightarrow \mu$, so the bound collapses to $|h(\mu)|$ and Proposition 6(iv) is recovered. As $\vartheta \rightarrow 0$ the electorate splits between viewers of only entertainment and viewers of only news, neither of whom any technology can move, and $|\Delta| \rightarrow 0$.

Figure 2 illustrates the intuition from Proposition 6(iii). What determines the fall in scrutiny is the convexity. Since Δ is the population average of an unchanged function h , and h bends, the degenerate distribution at μ is the lowest point available: replacing one voter at μ by two raises the average toward zero. Heterogeneity is thus a form of insurance against a substantial drop in scrutiny.

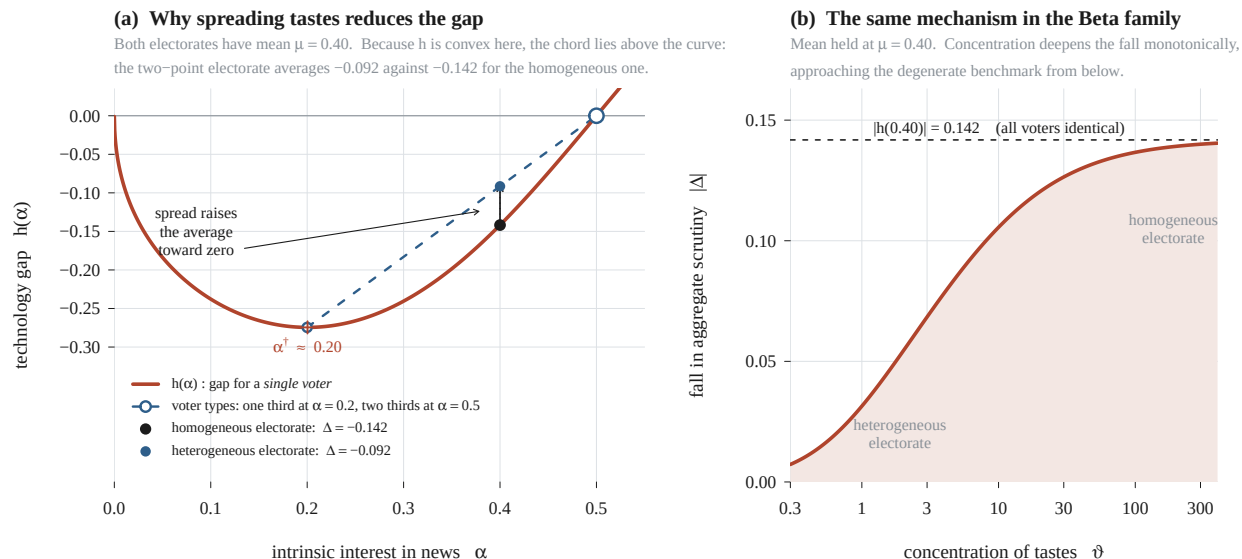


Figure 2: Why homogeneity widens the gap. Panel (a): the red curve is $h(\alpha)$, the gap faced by a single voter of taste α ; it is the same function for both electorates. The two electorates are the filled dots, both at their common mean taste $\mu = 0.40$. The homogeneous electorate lies *on* the curve, because $\mathbb{E}[h(\alpha)] = h(\mu)$ when tastes are degenerate; the electorate placing one third of voters at $\alpha = 0.20$ and two thirds at $\alpha = 0.50$ lies on the chord, which strict convexity puts above the curve. The vertical distance, 0.050, is the reduction in the fall of aggregate scrutiny gained by heterogeneity. Panel (b): the same impact of homogeneity within the Beta family at the same mean, where $|\Delta|$ rises with concentration towards the degenerate benchmark $|h(0.40)| = 0.142$. Parameters used in the benchmark: $\beta = \frac{1}{2}$, $\kappa_B = 2$, $\kappa_U = \frac{1}{2}$.

Corollary 9 (Political consequences). *Suppose $\mu < \frac{1}{2}$, so $\Delta < 0$ and aggregate scrutiny falls from $\tilde{t}_n$ to $\tilde{t}_n - |\Delta|$. Then expected first-period effort falls by $|\Delta|(1 - \frac{\tilde{t}_n}{2}) + O(\Delta^2) > 0$, and a good incumbent's re-election probability falls by $|\Delta|(1 - \rho_0)$. Unbundling, therefore, weakens discipline and selection together whenever the electorate prefers entertainment.*

A homogeneous, entertainment-leaning electorate might sharply reduce its news consumption once media transitions from traditional to digital, unbundling news from entertainment. Corollary 9 describes consequences of unbundling for accountability. In the next section, I discuss how much scrutiny — defined as news consumption — is optimal, and how it prevents a bad incumbent's pandering behavior.

6 How much scrutiny is optimal?

An incumbent takes aggregate news consumption $\tilde{t}_n \in [0, 1]$ as given. A good incumbent ($\theta = g$) faces zero effort cost and therefore exerts full effort in the first period, $e_1 = 1$, so that

$$\rho_g(\tilde{t}_n) = \tilde{t}_n. \quad (24)$$

A bad incumbent ($\theta = b$) draws an effort cost $c \sim U(0, 1)$ and, since no re-election incentive survives into the second period, chooses first-period effort by backward induction:

$$e_1^*(c; \tilde{t}_n) = \arg \max_{e \in [0, 1]} \tilde{t}_n \sqrt{e} - ce = \min \left\{ 1, \left(\frac{\tilde{t}_n}{2c} \right)^2 \right\}. \quad (25)$$

Effort is strictly positive for every c whenever $\tilde{t}_n > 0$, and reaches the corner $e_1^* = 1$ exactly when $c \leq \tilde{t}_n/2$. Since costs are uniform, the share of bad types exerting full effort is $\min\{1, \tilde{t}_n/2\}$: it rises linearly in scrutiny, reaching one half at $\tilde{t}_n = 1$. Averaging over the cost draw,

$$\mathbb{E}_c[e_1^*(\tilde{t}_n)] = \tilde{t}_n - \frac{\tilde{t}_n^2}{4}, \quad (26)$$

which is strictly increasing on $[0, 1]$, with slope $1 - \tilde{t}_n/2$ falling from 1 to $\frac{1}{2}$. Figure 3 plots (26) together with the individual best responses.

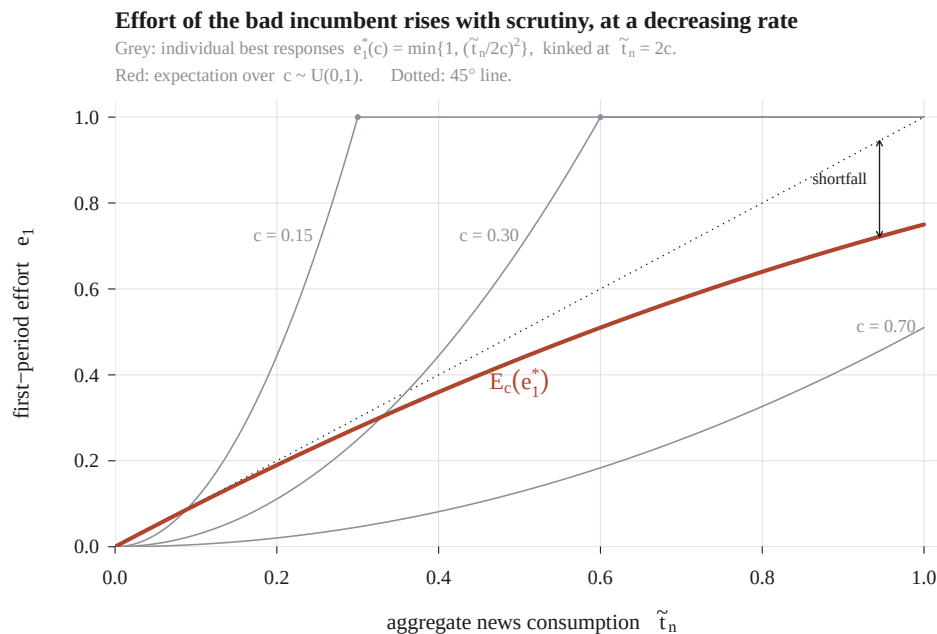


Figure 3: Best response of the bad incumbent. Grey curves are individual best responses $e_1^*(c; \tilde{t}_n)$ for selected cost draws, each kinked where $\tilde{t}_n = 2c$ and the corner is reached. The red curve is the expectation over $c \sim U(0, 1)$.

Discipline, therefore, increases monotonically in scrutiny. What is not monotone is how much the election reveals. The re-election probability of a bad type with cost draw c is $\rho_b(\tilde{t}_n; c) = \tilde{t}_n \sqrt{e_1^*(c; \tilde{t}_n)}$, and integrating over the cost distribution,

$$\mathbb{E}_c[\rho_b(\tilde{t}_n)] = \frac{\tilde{t}_n^2}{2} \left(1 - \ln \frac{\tilde{t}_n}{2} \right). \quad (27)$$

Figure 4(a) plots (27) against $\rho_g(\tilde{t}_n) = \tilde{t}_n$, which is the probability of reelection of a good incumbent. The fixed-cost curve traces a single representative bad type with $c = 0.5$, whose probability is $\rho_b = \tilde{t}_n^2$ until $\tilde{t}_n = 1$; the expected curve is the ex-ante probability voters face before the cost realization. The selection gap is the difference between the probability of reelection of a good incumbent and the ex-ante probability of reelection of a bad incumbent.

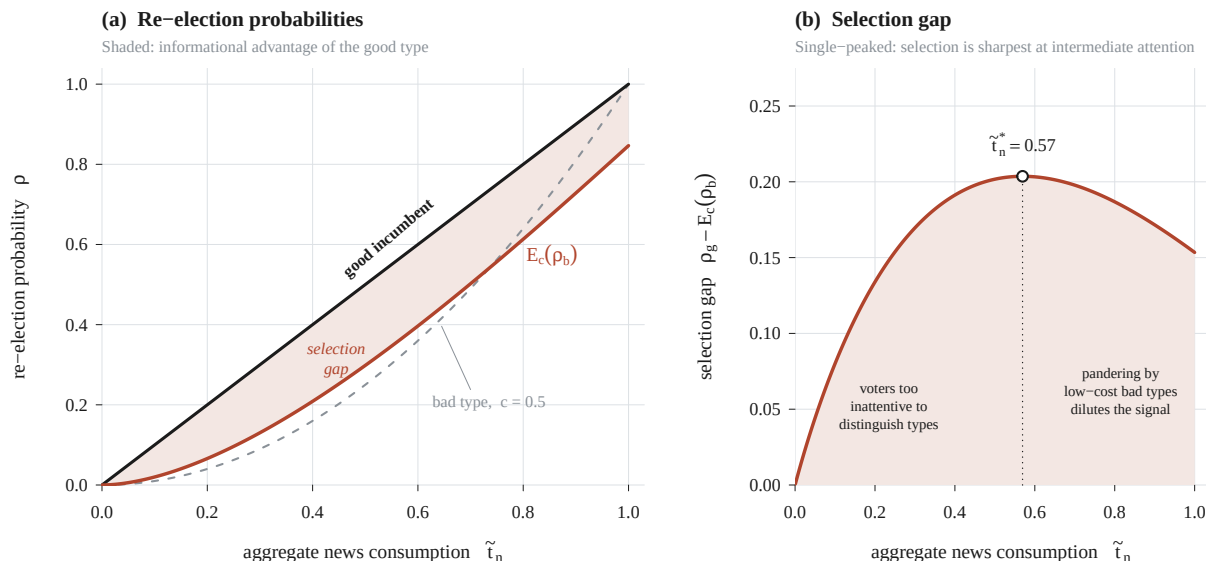


Figure 4: Re-election probabilities and the selection gap. Panel (a): probabilities by type. Panel (b): the gap $\rho_g - \mathbb{E}_c[\rho_b]$, single-peaked at $\tilde{t}_n^* \approx 0.57$.

The fixed-cost curve isolates the strategic behaviour of an individual bad type. Whenever $c < \tilde{t}_n/2$ the incumbent exerts full effort and becomes observationally indistinguishable from a good type. Such incumbents are low-cost relative to the monitoring environment: their absolute cost may be moderate, but heightened scrutiny makes imitation optimal. As $\tilde{t}_n$ rises, the set of bad types for whom this is true expands, and by $\tilde{t}_n = 1$ it comprises half of them.

Discipline and selection thus move apart. Greater attention always raises effort, so a bad incumbent under intense scrutiny delivers more first-period public good than one under weak scrutiny. But that same response makes the two types harder to tell apart, and the electorate is more likely to retain a bad politician who was merely imitating. When monitoring is weak, voters are too inattentive to distinguish types at all; when it is very strong, the informational advantage of the good type erodes through pooling. Electoral selection is strongest in an intermediate range — a “Goldilocks” region of voter attention. This mechanism is consistent with [Prato and Wolton \(2016\)](#), who show that excessive voter interest can encourage low-quality politicians to pander. This result is not specific to the functional form of the probability of re-election ρ - [Appendix H](#) illustrates the general case.

A corollary for welfare follows: stronger monitoring is not unambiguously desirable. It buys effort in the first period at the cost of a noisier signal for the second, and which effect dominates depends on how the second-period payoff weighs competence against realized output. I discuss this trade-off in [Section 7](#).

7 Voters’ welfare

[Proposition 6](#) establishes when unbundling lowers aggregate scrutiny. Separating news from entertainment does not necessarily lower welfare: voters value content as well as accountability, and the change in technology influences both. This section separates the two channels and shows that they move against each other exactly when the electorate leans towards entertainment.

Recall from Section 4 that a voter with taste α facing steering price κ attains indirect utility

$$v(\alpha; \kappa, \beta) = \kappa \ln \left[\beta \alpha^{1/\kappa} + (1 - \beta)(1 - \alpha)^{1/\kappa} \right], \quad (28)$$

the logarithm of the weighted power mean of $(\alpha, 1 - \alpha)$ of order $1/\kappa$ with weights $(\beta, 1 - \beta)$. Welfare per voter is the sum of private content utility and the non-rival accountability payoff,

$$W(\kappa; F) = \mathbb{E}_F[v(\alpha; \kappa, \beta)] + g(\tilde{t}_n(\kappa; F)), \quad \tilde{t}_n(\kappa; F) = \mathbb{E}_F[t_n^*(\alpha; \kappa, \beta)]. \quad (29)$$

Here v is the value of the voter's content problem alone — the maximum of content value net of steering cost — so the public good enters (29) only through the second term, and nothing is counted twice.

Writing $\tilde{t}_n^B = \tilde{t}_n(\kappa_B; F)$ for the average diet in the bundled environment, moving to the unbundled environment changes welfare by

$$\Delta W = \underbrace{G(F)}_{\text{private gain}} + \underbrace{g(\tilde{t}_n^B + \Delta(F)) - g(\tilde{t}_n^B)}_{\text{public effect}}, \quad G(F) \equiv \mathbb{E}_F[v(\alpha; \kappa_U, \beta) - v(\alpha; \kappa_B, \beta)]. \quad (30)$$

From now on, G , Δ and the critical stake S introduced in Proposition 10(iii) are evaluated in the large-electorate limit, where the civic term $\zeta = O(1/N)$ vanishes and the two environments differ only through κ . Voters do internalise the public good when choosing their diets — that is the term $\zeta = g'(\tilde{t}_n)/(N\kappa)$ in (18) — but only at rate $1/N$, so in the limit v is exactly the utility they attain.⁵

Proposition 10 (Welfare effects of unbundling). *Let $\kappa_B > \kappa_U$ and let g be increasing and concave. Unless specified, statements hold for any default bundle β .*

- (i) **Private Pareto improvement.** *For every voter with $\alpha \neq \frac{1}{2}$, $v(\alpha; \kappa_U) > v(\alpha; \kappa_B)$; voters with $\alpha = \frac{1}{2}$ are unaffected. Hence $G(F) > 0$ for every F other than the point mass at $\frac{1}{2}$. If $\beta = \frac{1}{2}$ the gain is symmetric, $v(1 - \alpha; \kappa) = v(\alpha; \kappa)$.*
- (ii) **Balanced or news-leaning electorates always gain.** *If F is symmetric about $\frac{1}{2}$ — in particular if $\mu = \frac{1}{2}$, for any ϑ — then $\Delta(F) = 0$ and $\Delta W = G(F) > 0$. More generally $\Delta W > 0$ whenever $\Delta(F) \geq 0$, which by Proposition 6(ii) holds in the Beta family for every $\mu \geq \frac{1}{2}$.*
- (iii) **Threshold on the political stake.** *Write $\tilde{t}_n^B = \tilde{t}_n(\kappa_B; F)$ and suppose $\Delta(F) < 0$. Unbundling reduces welfare if and only if the loss of accountability exceeds the private gain,*

$$g(\tilde{t}_n^B) - g(\tilde{t}_n^B + \Delta(F)) > G(F).$$

Equivalently, defining the critical stake

$$S(F) \equiv \frac{G(F)}{|\Delta(F)|} \in (0, \infty),$$

welfare falls if $g'(\tilde{t}_n^B) > S(F)$, and only if $g'(\tilde{t}_n^B + \Delta(F)) > S(F)$.

Proof in the Appendix G.

⁵In an electorate of finite size, the civic term is strictly positive, so a voter consumes slightly more news than the content-only optimum at which v is evaluated. However, this deviation costs her only a second order in ζ , that is $O(1/N^2)$.

Proposition 10 has the following consequences:

Nobody resists unbundling. By part (i) every voter is strictly better off in the unbundled environment, so the transition faces no private opposition — and yet, by part (iii), it can lower welfare. This is the model’s central externality. Each voter internalizes the full gain from a diet closer to her taste, but only $1/N$ of the effect of her attention on accountability, so the individually optimal response to cheaper steering is collectively excessive.

Homogeneity of preferences is the aggravating factor. Concentration of tastes simultaneously deepens the scrutiny loss (Proposition 6(iii)) and shrinks the private gain, since voters bunched near the mean are the ones the technology moves without compensating them much in content terms. Therefore, unbundling is most damaging exactly where the electorate is uniform and mildly disengaged, and least damaging where tastes are polarised.

Panel a) in Figure 5 illustrates private Pareto improvement after unbundling (except perfectly balanced voters at $\alpha = \frac{1}{2}$), with the largest gains for polarized voters. Regardless, total welfare might fall given the fall in accountability, as panel b) shows. Holding average interest in news fixed, μ , if voters are not sufficiently dispersed in their interest in news, the public loss might outweigh the gain in consuming more entertainment.

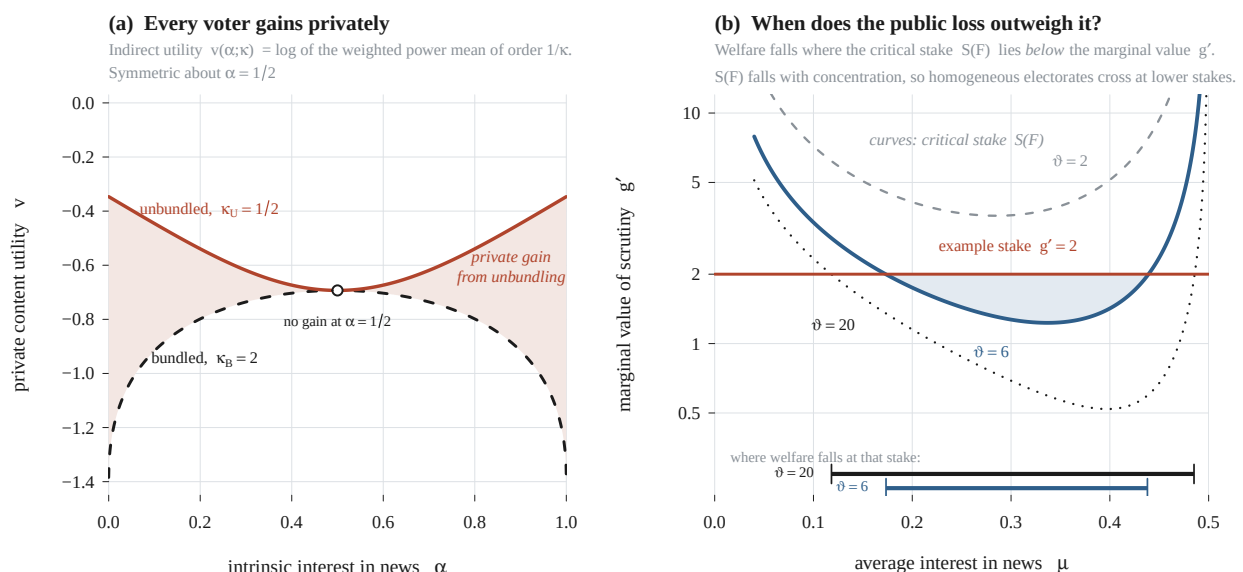


Figure 5: Welfare effects of unbundling. Panel (a): private content utility under the two technologies; the shaded region is the gain, strictly positive except at $\alpha = \frac{1}{2}$. Panel (b): the critical stake $S(F) = G(F)/|\Delta(F)|$ for three levels of taste concentration. Welfare falls wherever $S(F)$ lies below the marginal value of scrutiny g' ; the horizontal line marks the illustrative value $g' = 2$, and the bars beneath show the resulting range of average tastes over which unbundling is harmful. Since $S(F)$ falls with concentration, homogeneous electorates are harmed at lower political stakes and over a wider range of means. Parameters used: $\kappa_B = 2$, $\kappa_U = \frac{1}{2}$, $\beta = \frac{1}{2}$; g' is not a model primitive but the object against which $S(F)$ is compared, so no accountability technology is assumed.

8 Conclusion

This paper asks what happens to political accountability when exposure to political information stops being incidental and becomes a matter of choice. Modelling media technology as the price of steering attention away from whatever mixture a medium supplies by default,

I show that lowering that price can reduce aggregate scrutiny and weaken an incumbent’s incentive to exert effort — while making every individual voter better off.

The argument runs in three steps. Unbundling redistributes attention: entertainment-leaning voters withdraw from news and news-leaning voters specialise into it, with the biggest changes at intermediate tastes. Because those responses are mirror images, they cancel in a balanced electorate, and the aggregate moves only with the skew of tastes — falling when average interest in news lies below the midpoint, and falling most where tastes are homogeneous. Aggregate scrutiny then does two things to elections: it always raises incumbent effort, but it also makes types harder to tell apart, as low-cost bad incumbents imitate good ones. Moreover, every voter gains privately from the change, so the whole welfare cost falls on a public good that no one internalizes.

What the mechanism does *not* require is worth stating, because the standard explanations are all absent. There is no misinformation: news is accurate and freely available wherever it is produced. There is no manipulation: media firms are demand-followers, with no power over the composition of what voters consume and no influence beyond the discrete decision of whether reporting exists at all. There is no irrationality: voters correctly anticipate the incumbent’s effort and optimise given it. And there is no sorting into ideological camps: tastes here are over content type, not over political position, and the model contains no partisanship. Accountability weakens regardless.

These findings rationalize the contemporary media landscape. According to the 2026 Reuters Institute Digital News Report, news consumption via traditional channels such as TV, newspapers, and “owned and operated” news websites has been declining for years across 48 surveyed countries. At the same time, the interest in news has also been declining: since 2021, the proportion of people who are interested in news has fallen by 13 percentage points (Egan et al. 2026). Two phenomena have therefore moved together: news has become easier to separate from entertainment, and the average taste for it has fallen. The model implies that these compounds do not offset each other. The consequences are most acute locally. Investigative journalism is costlier than the content that substitutes for it — reprinted wire stories, listings, weather — and with the internet supplying that content directly, many localities risk falling below the critical mass of attention that sustains a newsroom at all.⁶

In this context, policy implications are focused on the demand. Interventions that lower the cost of producing news relax the critical mass and can restore local provision, but they do not change how much news voters choose to consume once it exists. Instead, measures that raise the value voters place on being informed would improve accountability in this model. Especially in relatively small constituencies, where the electorate is uniform in taste for news and mildly disengaged.

There are several natural extensions to this model. The default mixture is exogenous here, and endogenizing it is the obvious next step. The interesting extension would couple the choice of default with a participation decision, letting viewers abandon a media firm whose default sits too far from their taste, and letting platforms personalize the default mix as recommender systems do. Moreover, making the taste for news depend on past consumption would close a feedback loop in which unbundling lowers news consumption, which lowers

⁶Local news: How publishers can still provide value in a platform world by Richard Fletcher, June 2025. Access: 9/7/2026.

the taste for news, which lowers consumption again. A third direction merges the welfare analysis with the extensive margin. Proposition 1 identifies the markets in which unbundling extinguishes reporting altogether. Voters there are rationed and lose content value as well as scrutiny, which would give welfare a discontinuity at the critical mass and a distinct policy case for sustaining marginal newsrooms. Finally, Appendix B lets voters differ in how much they value the public good but treats that valuation as independent of content taste. Correlating the two — plausibly, those who prefer entertainment also care less about local government — would make the two channels add up rather than average out.

The model also delivers comparative statics that are in principle measurable. Tastes enter only through two parameters, the mean μ and the dispersion, denoted by ϑ . Proposition 6(ii) predicts that unbundling lowers aggregate scrutiny if and only if $\mu < \frac{1}{2}$; Proposition 6(iii) and Corollary 7 predict that the fall is larger the more concentrated tastes are at a given mean, bounded above by the degenerate benchmark $|h(\mu)|$. Neither requires tastes to be observed directly, since they can be recovered from media diets by inverting (18). Testing the two predictions against local-government outcomes would be a natural empirical companion to this paper.

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A Micro-foundation: probabilistic voting

I present ρ as a reduced form of the probability of re-election, restricted by economically meaningful properties, following political economy literature (Besley and Prat (2006), Strömberg and Prat (2005), Ferejohn (1986)). This subsection presents one election that generates the used reduced form. The qualitative results do not depend on that particular form: Appendix H shows that the disciplining and pandering results hold for any re-election technology satisfying the monotonicity and curvature conditions stated here. What the reduced form does determine are the magnitudes — the value of ρ_0 (probability of re-election if nobody consumes news), the marginal social value of scrutiny g' (derivative of the value of public good in a voter's utility function with respect to $t_{n,J}$), and hence the threshold of Proposition 10.

The electorate. I assume that a voter's average demand for news is proportional to the probability that he registers information about the incumbent's first-period conduct. A share $\tilde{t}_n = \frac{1}{N} \sum_J t_J$ of the electorate is therefore informed, and the remainder uninformed.

Information and beliefs. An informed voter observes a binary verdict, *pass* or *fail*. An incumbent exerting effort e passes with probability $p(e)$, where p is increasing and concave with $p(0) = 0$ and $p(1) = 1$; a good type exerts $e_1 = 1$ and passes with certainty. If e_b is the effort of a bad type, Bayes' rule gives

$$\Pr(g \mid \text{pass}) = \frac{\gamma}{\gamma + (1 - \gamma)p(e_b)} > \gamma, \quad \Pr(g \mid \text{fail}) = 0, \quad (31)$$

the latter because only bad types fail. Uninformed voters hold the prior γ .

Voting. Re-electing the incumbent delivers τ_2 if she is good and nothing if she is bad, since a bad type shirks in her last period; replacing her delivers a first-term official, good with prior probability γ , worth τ_1 . Define the *informational advantage* of the incumbent in the eyes of a voter with information set $\mathcal{I} = \{\text{pass}, \text{fail}\}$,

$$\Delta(\mathcal{I}) = \Pr(g \mid \mathcal{I}) \tau_2 - \gamma \tau_1, \quad (32)$$

which is the expected gain in the second-period public good from keeping the incumbent rather than drawing a fresh first-term official. It has two features: the tenure premium $\tau_2 > \tau_1$, which favours the incumbent regardless of what voters learn, and information, which favours her after a favourable verdict and turns against her after an unfavourable one. Informational advantage of uninformed voter is denoted as Δ_U , and of informed voter as Δ_I . We thus have:

$$\Delta_U = \gamma(\tau_2 - \tau_1) > 0, \quad \Delta_I(e_1) = \Pr(g \mid \text{pass}) \tau_2 - \gamma \tau_1, \quad (33)$$

the latter increasing in the incumbent's effort. As is standard in probabilistic voting models, voters also differ in idiosyncratic, non-policy assessments of the two candidates ψ_J with distribution H_ψ , and the electorate is subject to an aggregate popularity shock ξ with distribution G_ξ .

From individual votes to the reduced form. Voter J backs the incumbent when her advantage, adjusted by her idiosyncratic assessment and the common shock, is positive: $\Delta(\mathcal{I}_J) + \psi_J + \xi \geq 0$. Voters within an information group share the same Δ and the same ξ and differ only in ψ_J , so the share of group $i \in \{I, U\}$ backing the incumbent is $H_{\psi_i}(\Delta_i + \xi)$, and vote share for her is

$$VS = \tilde{t}_n H(\Delta_I(e_1) + \xi) + (1 - \tilde{t}_n) H(\Delta_U + \xi).$$

Suppose that ψ_J is distributed uniformly on $[-A, A]$ with $A > 0$, and $z = \Delta_i + \xi$. Then, the share of information group $i \in \{I, U\}$ can be expressed as $H(z) = \frac{1}{2} + z/(2A)$ on $[-A, A]$. Conditional on the realized popularity shock, $\Delta_i + \xi$ is a number, and the only variation across voters within group i is the idiosyncratic term, so the group's realized support is $H_{\psi}(\Delta_i + \xi)$, where H is the distribution of ψ_J . The two disturbances, therefore, enter at different levels: idiosyncratic assessments average out within each group and determine its vote share, while the common shock does not average out and is what makes the electoral outcome uncertain.

I assume A is large enough that $\Delta_j + \xi$ lies in $[-A, A]$ for both groups and all realizations of ξ . Outside this range, H is flat at 0 or 1 and the group votes unanimously, so its support would not respond to its beliefs at all; inside it, $H' = 1/(2A) > 0$ and a marginal improvement in a group's assessment of the incumbent converts into a marginal gain in her vote share.

The incumbent's vote share is then

$$VS = \tilde{t}_n \left[\frac{1}{2} + \frac{\Delta_I(e_1) + \xi}{2A} \right] + (1 - \tilde{t}_n) \left[\frac{1}{2} + \frac{\Delta_U + \xi}{2A} \right] = \frac{1}{2} + \frac{\tilde{t}_n \Delta_I(e_1) + (1 - \tilde{t}_n) \Delta_U + \xi}{2A}$$

Under majority rule, the incumbent is therefore retained exactly when

$$\tilde{t}_n \Delta_I(e_1) + (1 - \tilde{t}_n) \Delta_U + \xi > 0. \quad (34)$$

Hence

$$\rho(\tilde{t}_n, e_1) = G_\xi \left(\tilde{t}_n \Delta_I(e_1) + (1 - \tilde{t}_n) \Delta_U \right). \quad (35)$$

Suppose $\xi \sim U[a, b]$, with density $\kappa \equiv 1/(b - a) > 0$. Over the support, G_ξ is affine,

$$G_\xi(x) = \frac{x - a}{b - a} = \kappa x + c, \quad c \equiv -\kappa a. \quad (36)$$

Then, we can rewrite (35) as:

$$\begin{aligned} G_\xi(\tilde{t}_n \Delta_I + (1 - \tilde{t}_n) \Delta_U) &= \kappa [\tilde{t}_n \Delta_I + (1 - \tilde{t}_n) \Delta_U] + c \\ &= \tilde{t}_n \kappa \Delta_I + (1 - \tilde{t}_n) \kappa \Delta_U + [\tilde{t}_n + (1 - \tilde{t}_n)] c \\ &= \tilde{t}_n (\kappa \Delta_I + c) + (1 - \tilde{t}_n) (\kappa \Delta_U + c) \\ &= \tilde{t}_n G_\xi(\Delta_I(e_1)) + (1 - \tilde{t}_n) G_\xi(\Delta_U). \end{aligned} \quad (37)$$

The intercept c cancels out because the weights sum to one. Defining

$$q(e_1) \equiv G_\xi(\Delta_I(e_1)), \quad \rho_0 \equiv G_\xi(\Delta_U) = G_\xi(\gamma(\tau_2 - \tau_1)), \quad (38)$$

so we can rewrite (35) as:

$$\rho(\tilde{t}_n, e_1) = \tilde{t}_n q(e_1) + (1 - \tilde{t}_n) \rho_0. \quad (39)$$

Evaluating (39) at $\tilde{t}_n = 0$ returns ρ_0 and at $\tilde{t}_n = 1$ returns $q(e_1)$; in particular $\rho(0, e_1)$ does not depend on e_1 , as an electorate that does not observe anything cannot condition on perceived effort.

From (36) and (38) we have $q(e_1) = \kappa \Delta_I(e_1) + c$. Decomposing the informed voters' differential across the two signal realizations,

$$\Delta_I(e_1) = p(e_1)\Delta_{\text{pass}} + (1 - p(e_1))\Delta_{\text{fail}} = \Delta_{\text{fail}} + p(e_1)(\Delta_{\text{pass}} - \Delta_{\text{fail}}), \quad (40)$$

so that

$$q'(e_1) = \kappa p'(e_1)(\Delta_{\text{pass}} - \Delta_{\text{fail}}) > 0, \quad q''(e_1) = \kappa p''(e_1)(\Delta_{\text{pass}} - \Delta_{\text{fail}}) < 0, \quad (41)$$

where the signs follow from $\Delta_{\text{pass}} > \Delta_{\text{fail}}$, itself implied by $\Pr(g \mid \text{pass}) > \gamma > 0 = \Pr(g \mid \text{fail})$. Hence q inherits monotonicity and concavity from p through the positive constant $\kappa(\Delta_{\text{pass}} - \Delta_{\text{fail}})$. The specification $q(e_1) = \sqrt{e_1}$ corresponds to $p(e_1) = \sqrt{e_1}$ with this constant normalised to one and the intercept to zero.

B Extension: heterogenous valuation of the public good

The main text places weight one on the public good for every voter. This appendix relaxes this restriction: voters value the public good at $\lambda_J > 0$. The main text is the special case $\lambda_J \equiv 1$: a voter with $\lambda_J < 1$ values governance quality *less* than the baseline citizen, one with $\lambda_J > 1$ values it *more*. We write $\bar{\lambda} = \frac{1}{N} \sum_J \lambda_J$ for the population mean, so that $\bar{\lambda} \gtrless 1$ says whether the electorate values governance more or less than the main text assumes.

Setup and demand Let voter J place private valuation $\lambda_J \geq 0$ on the public good, so that her objective becomes

$$\max_{t \in (0,1)} t \ln \alpha_J + (1 - t) \ln(1 - \alpha_J) - C_\kappa(t) + \lambda_J g(\tilde{t}_n), \quad \tilde{t}_n = \frac{1}{N} \sum_I t_I. \quad (42)$$

The public good remains nonrival — every voter enjoys $g(\tilde{t}_n)$ undivided — and a voter's own attention moves the aggregate only at rate $1/N$. Differentiating (42) and using $C'_\kappa(t) = \kappa \ln \frac{t}{1-t}$, the first-order condition is

$$\kappa \ln \frac{t}{1-t} = x_J + \frac{\lambda_J g'(\tilde{t}_n)}{N}, \quad x_J = \ln \frac{\alpha_J}{1 - \alpha_J},$$

which inverts to

$$t^*(\alpha_J, \lambda_J; \kappa) = \frac{\alpha_J^{1/\kappa} e^{\zeta_J}}{\alpha_J^{1/\kappa} e^{\zeta_J} + (1 - \alpha_J)^{1/\kappa}}, \quad \boxed{\zeta_J \equiv \frac{\lambda_J g'(\tilde{t}_n)}{N\kappa}} \quad (43)$$

where ζ_J is voter J 's *civic term*.

Two features of (43) organize the analysis.

Proposition 11 (Civic motives require unbundling). *The civic term $\zeta_J = \lambda_J g'(\tilde{t}_n)/(N\kappa)$ is decreasing in the stiffness κ . Civic concern, therefore, translates into news consumption more strongly in unbundled environments: for given λ_J , halving κ doubles the term.*

Unbundling lets tastes α_J express themselves, which withdraws the entertainment-lovers from the news audience, but it equally lets civic motives λ_J express themselves. A society with a strong civic culture is correspondingly less exposed to the accountability cost of unbundling.

Proposition 12 (Two roles of the civic valuation). *The civic valuation λ_J enters the model twice, at different orders:*

- (i) **Behavior (order $1/N$).** *Voter J 's diet has larger share of news by $\zeta_J = \lambda_J g'(\tilde{t}_n)/(N\kappa)$. The effect vanishes as $N \rightarrow \infty$.*
- (ii) **Welfare (order 1).** *The social cost of a fall Δ in scrutiny is $\bar{\lambda}[g(\tilde{t}_n) - g(\tilde{t}_n - \Delta)]$, which does not vanish with N .*

Hence a change in civic valuations alters what unbundling is worth far more than what citizens do about it.

This asymmetry is the free-rider problem. Citizens who value governance twice as much, consume only marginally more news — their influence on the aggregate is diluted by N either way — yet they suffer twice the loss when scrutiny falls. Civic valuation is therefore almost entirely a welfare parameter, not a behavioral one, and a society cannot rely on citizens caring more to protect its own monitoring.

C Equilibrium

Additional notation. Let $N \in \mathbb{N}$ denote the number of voters. Voter J has taste $\alpha_J \in (0, 1)$ and, given the media environment (κ, β) and aggregate scrutiny, a desired news share $t_{J,n}^* \in (0, 1)$. Let $\nu \in \{0, 1\}$ indicate whether news is produced. Since news is supplied only by the modeled firms, and is unconstrained where a newsroom exists, realized consumption is

$$\tilde{t}_{J,n} := \nu t_{J,n}^*, \quad \tilde{t}_{J,e} := 1 - \tilde{t}_J,$$

and average desired and realized news consumption are

$$t_n^* := \frac{1}{N} \sum_{J=1}^N t_{J,n}^*, \quad \tilde{t}_n := \frac{1}{N} \sum_{J=1}^N \tilde{t}_{J,n} = \nu t_n^*.$$

Denote $x_J \equiv \ln \frac{\alpha_J}{1-\alpha_J}$ for voter J 's taste log-odds and $\bar{b} \equiv \ln \frac{\beta}{1-\beta}$ for the log-odds of the environment's default mix, so that the marginal steering cost is $C'_{\kappa,\beta}(t_{J,n}) = \kappa \left[\ln \frac{t_{J,n}}{1-t_{J,n}} - \bar{b} \right]$.

Media profits. Advertising revenue is proportional to attention devoted to each content type on the platform, at rates $A_n > A_e > 0$. A platform pays a fixed cost FC_e to operate and, if it funds a newsroom, an additional fixed cost FC_n — while news carries no marginal cost once the newsroom exists. With $M \in \mathbb{N}$ symmetric platforms sharing the market, per-firm profit given the newsroom decision ν is

$$\Pi(\nu; t_n^*, M) = \frac{N}{M} \left[A_n \nu t_n^* + A_e (1 - \nu t_n^*) \right] - FC_e - \nu FC_n, \quad (44)$$

or equivalently

$$\Pi(\nu; t_n^*, M) = \frac{N}{M} \left[A_e + \nu (A_n - A_e) t_n^* \right] - FC_e - \nu FC_n. \quad (45)$$

Entertainment revenue A_e is gained on all attention regardless of ν . Funding a newsroom gives the premium $A_n - A_e$ on the share of attention devoted to the news, t_n^* .

Definition. An equilibrium is a profile

$$\left((t_{J,n}^*)_{J=1}^N, (e_1^\theta, e_2^\theta)_{\theta \in \{g,b\}}, \nu, M, \tilde{t}_n, \rho \right)$$

such that:

- (i) **Consumers.** Each voter J chooses $t_{J,n}^* \in (0, 1)$ to maximize content value net of steering cost, internalizing the effect of her own attention on the public good:

$$t_{J,n}^* \in \arg \max_{t_{J,n} \in (0,1)} \left[t_{J,n} \ln \alpha_J + (1 - t_{J,n}) \ln(1 - \alpha_J) \right] - C_{\kappa,\beta}(t) + g\left(\frac{t_{J,n} + \sum_{I \neq J} \tilde{t}_{I,n}}{N}\right),$$

taking the diets of others as given. Equivalently, $t_{J,n}$ satisfies

$$\ln \frac{t_{J,n}}{1 - t_{J,n}} = \frac{x_J}{\kappa} + \bar{b} + \zeta(\tilde{t}_n), \quad \zeta(\tilde{t}_n) \equiv \frac{g'(\tilde{t}_n)}{N\kappa}, \quad x_J = \ln \frac{\alpha_J}{1 - \alpha_J}. \quad (46)$$

- (ii) **Media firms (newsroom and entry).** Given the demand a newsroom would attract, firms choose (ν, M) :

- (a) *Newsroom optimality:* $\nu \in \arg \max_{u \in \{0,1\}} \Pi(u; t_n^*, M)$, i.e. $\nu = 1$ if and only if $\frac{N}{M}(A_n - A_e)t_n^* \geq FC_n$;
- (b) *Free entry:* if $M \geq 1$ then $\Pi(\nu; t_n^*, M) \geq 0$ and $\Pi(\nu; t_n^*, M + 1) < 0$.

If $\max_{u \in \{0,1\}} \Pi(u; t_n^*, 1) < 0$ then $M = 0$ and $\nu = 0$.

- (iii) **Incumbent.** Given realized average news $\tilde{t}_n$, the incumbent chooses effort by backward induction: a good type exerts $e_1^g = e_2^g = 1$; a bad type with cost c exerts $e_2^b = 0$ and $e_1^b(c) = \min\{1, (\tilde{t}_n/2c)^2\}$.

- (iv) **Aggregation.** $\tilde{t}_n = \nu t_n^*$ with $t_n^* = \frac{1}{N} \sum_J t_{J,n}$.

- (v) **Consistency.** Beliefs are Bayesian on the equilibrium path; the marginal public good $g'(\tilde{t}_n)$ entering (46) is evaluated at the incumbent's equilibrium effort from (iii); and $\rho = \rho(\tilde{t}_n, e_1)$.

Existence and uniqueness. Conditions (i) and (iv) are simultaneous: individual diets depend on aggregate scrutiny through the civic term ζ , and aggregate scrutiny is the average of individual diets. Define, for a level of scrutiny $s \in [0, 1]$,

$$\Phi(s) := \frac{1}{N} \sum_{J=1}^N \Lambda\left(\frac{x_J}{\kappa} + \bar{b} + \zeta(s)\right), \quad \Lambda(z) = \frac{1}{1 + e^{-z}}, \quad (47)$$

the average diet that voters choose when they expect scrutiny s . An equilibrium with $\nu = 1$ requires $\tilde{t}_n = \Phi(\tilde{t}_n)$.

Proposition 13. *Assume g is continuously differentiable on $[0, 1]$ and twice continuously differentiable on $(0, 1]$. Then an equilibrium exists, and it is unique provided the feedback from aggregate scrutiny back to individual attention is weak:*

$$N\kappa > \frac{1}{4} \max |g''|, \quad (48)$$

the maximum being taken over the levels of scrutiny that can arise in equilibrium.

Reading the condition. There is a loop in the model: when voters expect more scrutiny, the marginal value of news changes — at rate g'' — which alters their diets, which in turn alters scrutiny. Condition (48) requires that this loop weakens rather than amplify, so that the process settles at a single point rather than running away. It is easily satisfied, for two reasons: each voter's influence on the aggregate is only $1/N$, and steering is costly, so diets respond sluggishly to any change in incentive. Because g is concave there, more monitoring by others lowers the marginal value of monitoring, so each voter responds by consuming slightly less news, so attention choices are strategic substitutes.

Proof. Existence and uniqueness of equilibrium scrutiny. Recall from (47) that

$$\Phi(s) = \frac{1}{N} \sum_{J=1}^N \Lambda\left(\frac{x_J}{\kappa} + \bar{b} + \zeta(s)\right), \quad \zeta(s) = \frac{g'(s)}{N\kappa}, \quad \Lambda(z) = \frac{1}{1 + e^{-z}},$$

is the average diet voters choose when they expect scrutiny s . An equilibrium with $\nu = 1$ is a level of scrutiny that reproduces itself, $\tilde{t}_n = \Phi(\tilde{t}_n)$.

Step 1: how the average diet responds to expected scrutiny. Differentiating Φ by the chain rule, and using the logistic identity $\Lambda'(z) = \Lambda(z)(1 - \Lambda(z))$ — so that the derivative evaluated at voter J 's argument is simply $t_{J,n}(1 - t_{J,n})$ — gives

$$\Phi'(s) = \underbrace{\frac{g''(s)}{N\kappa}}_{\zeta'(s)} \cdot \underbrace{\frac{1}{N} \sum_{J=1}^N t_{J,n}(1 - t_{J,n})}_{R(s)}, \quad (49)$$

the product of two terms. The first is how much a change in scrutiny alters the civic term: it is the curvature g'' of the public good, diluted by the electorate size N and by the price of steering κ . The second, $R(s)$, is the average responsiveness of diets to a change in incentive; since $t_{J,n} \in (0, 1)$ for every voter, $R(s) \in (0, \frac{1}{4}]$.

Step 2: the sign. Because $R(s) > 0$ and $N\kappa > 0$, the sign of $\Phi'(s)$ is the sign of $g''(s)$. Differentiating the public good twice,

$$g''(t_n) = (1 - \gamma)\tau_1 \left[\frac{\gamma-1}{2} + \gamma \ln \frac{t_n}{2} \right] < 0 \quad \text{for all } t_n \in (0, 1],$$

Hence $\Phi'(s) < 0$: *the more scrutiny voters expect, the less news each chooses to consume.* The public good is concave in monitoring, so additional scrutiny is worth progressively less at the margin. Attention choices are strategic substitutes.

Step 3: exactly one fixed point. Let $\psi(s) := \Phi(s) - s$ measure the gap between the scrutiny voters deliver and the scrutiny they expected. Then ψ is continuous, and strictly decreasing because $\psi'(s) = \Phi'(s) - 1 < -1 < 0$. At the two ends,

$$\psi(0) = \Phi(0) > 0, \quad \psi(1) = \Phi(1) - 1 < 0,$$

At the left endpoint, $\psi(0) = \Phi(0) > 0$: even voters who expect no scrutiny at all still consume news for its own sake, since the logistic Λ is strictly positive and the steering cost rules out a corner diet. At the right endpoint, $\psi(1) = \Phi(1) - 1 < 0$ for the symmetric reason, no voter consuming only news. Since ψ is continuous and changes sign, it has a zero $\tilde{t}_n^* \in (0, 1)$. Since Φ takes values strictly inside $(0, 1)$: every voter's diet is interior, so the average is neither 0 nor 1. A continuous, strictly decreasing function that is positive at one end of an interval and negative at the other has exactly one zero. That zero, $\tilde{t}_n^*$, is the

unique fixed point of Φ : the unique level of scrutiny that voters both expect and deliver. Geometrically, Φ is a downward-sloping best-response curve, and a downward-sloping curve crosses the 45° line once and only once.

Step 4: the remaining components. Given $t_n^* = \tilde{t}_n^*$, per-firm profit (8) is strictly decreasing in M , so the free-entry condition determines a unique number of platforms; given M , the newsroom condition determines ν uniquely, with the convention $\nu = 1$ on the case $\frac{N}{M}(A_n - A_e)t_n^* - FC_n$ where the firm is indifferent. This fixes $\tilde{t}_n = \nu \tilde{t}_n^*$. Given $\tilde{t}_n$, the good type's problem is trivial, and the bad type's objective is strictly concave in effort, so her effort is unique; ρ then follows from $(\tilde{t}_n, e_1)$. The equilibrium profile is therefore unique. $\square$

D Proof of Lemma 2

Notation: fix $\kappa > 0$, $\beta \in (0, 1)$, a taste $\alpha_J \in (0, 1)$ and a civic term $\zeta \in \mathbb{R}$, and set $q = 1 - \kappa$, so that $q < 1$.

Step 1: the CES first-order condition. Write the CES objective as $U(t) = G(t)^{1/q}$ with

$$G(t) = (1 - a)(1 - t)^q + a t^q > 0 \quad \text{for } t \in (0, 1).$$

Differentiating,

$$G'(t) = q a t^{q-1} + q(1 - a)(1 - t)^{q-1} \times (-1) = q \left[a t^{q-1} - (1 - a)(1 - t)^{q-1} \right],$$

and hence, by the chain rule,

$$U'(t) = \frac{1}{q} G(t)^{\frac{1}{q}-1} G'(t) = G(t)^{\frac{1}{q}-1} \left[a t^{q-1} - (1 - a)(1 - t)^{q-1} \right], \quad (50)$$

the factor q cancelling between the outer and inner derivatives. Since $G(t)^{\frac{1}{q}-1} > 0$ on $(0, 1)$, the sign of U' is the sign of the bracket

$$B(t) := a t^{q-1} - (1 - a)(1 - t)^{q-1}.$$

Step 2: the optimum is interior and unique. Differentiating B ,

$$B'(t) = (q - 1) \left[a t^{q-2} + (1 - a)(1 - t)^{q-2} \right] < 0 \quad \text{for all } t \in (0, 1),$$

because $q - 1 = -\kappa < 0$ while the bracketed term is strictly positive. So B is strictly decreasing. At the boundaries, $q - 1 < 0$ implies $t^{q-1} \rightarrow +\infty$ as $t \rightarrow 0^+$ and $(1 - t)^{q-1} \rightarrow +\infty$ as $t \rightarrow 1^-$, so

$$\lim_{t \rightarrow 0^+} B(t) = +\infty, \quad \lim_{t \rightarrow 1^-} B(t) = -\infty.$$

A strictly decreasing continuous function running from $+\infty$ to $-\infty$ has exactly one zero $t^\dagger \in (0, 1)$, and U' is positive below it and negative above it. Hence U has a unique interior maximizer, characterized by $B(t^\dagger) = 0$, i.e.

$$a t^{q-1} = (1 - a)(1 - t)^{q-1}. \quad (51)$$

Step 3: solving the CES condition. Rearranging (51),

$$\frac{a}{1-a} = \frac{(1-t)^{q-1}}{t^{q-1}} = \left(\frac{1-t}{t}\right)^{q-1} = \left(\frac{t}{1-t}\right)^{1-q},$$

Raising both sides to the power $1/(1-q) = 1/\kappa$,

$$\frac{t}{1-t} = \left(\frac{a}{1-a}\right)^{1/\kappa}. \quad (52)$$

Step 4: the steering-cost condition. Exponentiating (18) and using $e^{x_J} = \alpha_J/(1-\alpha_J)$ and $e^{\bar{b}} = \beta/(1-\beta)$,

$$\frac{t^*}{1-t^*} = e^{x_J/\kappa} e^{\bar{b}} e^{\zeta} = \left(\frac{\alpha_J}{1-\alpha_J}\right)^{1/\kappa} \frac{\beta e^{\zeta}}{1-\beta}. \quad (53)$$

Step 5: matching the two. Equating (52) and (53),

$$\left(\frac{a}{1-a}\right)^{1/\kappa} = \left(\frac{\alpha_J}{1-\alpha_J}\right)^{1/\kappa} \frac{\beta e^{\zeta}}{1-\beta},$$

and raising both sides to the power κ gives (21):

$$\frac{a}{1-a} = \frac{\alpha_J}{1-\alpha_J} \left(\frac{\beta e^{\zeta}}{1-\beta}\right)^{\kappa}.$$

□

Remark 14 (The Cobb–Douglas boundary). At $\kappa = 1$ we have $q = 0$, where $G^{1/q}$ is undefined and the CES objective must be read as its limit, the Cobb–Douglas form $t^a(1-t)^{1-a}$. Its first-order condition $a/t = (1-a)/(1-t)$ gives $t^* = a$, consistent with (52) at $\kappa = 1$; with $\beta = \frac{1}{2}$ and $\zeta = 0$, (21) then yields $a = \alpha_J$ and hence $t^* = \alpha_J$, matching (19).

E Proof of Lemma 3

Denote $x = \ln \frac{\alpha}{1-\alpha}$ and $\bar{b} = \ln \frac{\beta}{1-\beta}$, and set $m \equiv 1/\kappa$. By (18) the optimal diet under technology κ is

$$s_m(\alpha) = \Lambda(mx + \bar{b}), \quad \Lambda(z) = \frac{1}{1 + e^{-z}},$$

so $t_n^*(\alpha; \kappa, \beta) = s_{1/\kappa}(\alpha)$. Since $\kappa_B > \kappa_U$ we have $m_U \equiv 1/\kappa_U > m_B \equiv 1/\kappa_B$, and $h = s_{m_U} - s_{m_B}$.

I start the proof from Lemma 3(ii):

(ii) Λ is strictly increasing, so $\partial s_m / \partial m = \Lambda'(mx + \bar{b}) x$ has the sign of x . For $\alpha < \frac{1}{2}$ we have $x < 0$, so s_m is strictly decreasing in m ; since $m_U > m_B$ this gives $h < 0$ on $(0, \frac{1}{2})$. For $\alpha > \frac{1}{2}$ we have $x > 0$, the monotonicity reverses, and $h > 0$. Neither step uses the value of $\bar{b}$, so both hold for every default β .

(i) As $\alpha \rightarrow 0$, $x \rightarrow -\infty$ and $s_m \rightarrow 0$ for every $m > 0$; as $\alpha \rightarrow 1$, $x \rightarrow +\infty$ and $s_m \rightarrow 1$. At $\alpha = \frac{1}{2}$ we have $x = 0$, hence $s_m = \Lambda(\bar{b}) = \beta$ for every m : both technologies deliver the default mix, and the gap closes. So h vanishes at 0, $\frac{1}{2}$ and 1, whatever the default. These are the only zeros, because by (ii) h has a strict sign in each of the two intervals in between.

(iii) In the log-odds parametrisation $s_m(\alpha; \beta) = \Lambda(mx + \bar{b})$ with $x = \ln \frac{\alpha}{1-\alpha}$ and $\bar{b} = \ln \frac{\beta}{1-\beta}$, the map $(\alpha, \beta) \mapsto (1-\alpha, 1-\beta)$ sends $x \mapsto -x$ and $\bar{b} \mapsto -\bar{b}$, hence $mx + \bar{b} \mapsto -(mx + \bar{b})$ for every m . Since $\Lambda(-z) = 1 - \Lambda(z)$,

$$s_m(1-\alpha; 1-\beta) = 1 - s_m(\alpha; \beta) \quad \text{for every } m > 0.$$

Differencing this identity across m_U and m_B , the constants cancel and $h(1-\alpha; 1-\beta) = -h(\alpha; \beta)$.

(iv) Let $\beta = \frac{1}{2}$, so $\bar{b} = 0$.

Step 1. With $\bar{b} = 0$ we have $s_m = \Lambda(mx)$. Replacing α by $1-\alpha$ replaces x by $-x$, and $\Lambda(-z) = 1 - \Lambda(z)$, so $s_m(1-\alpha) = 1 - s_m(\alpha)$ for every m , and therefore

$$h(1-\alpha) = -h(\alpha), \quad \alpha \in (0, 1).$$

It is enough to work on $(0, \frac{1}{2})$; statements on $(\frac{1}{2}, 1)$ follow by reflection.

Step 2. With $\bar{b} = 0$ write $s_m(\alpha) = \alpha^m / D_m$, $D_m = \alpha^m + (1-\alpha)^m$, and $P = \alpha(1-\alpha)$. Differentiating, the cross-terms cancel and

$$s'_m(\alpha) = \frac{m P^{m-1}}{D_m^2}, \quad s''_m(\alpha) = \frac{m P^{m-2}}{D_m^3} E_m(\alpha),$$

$$E_m(\alpha) = (m-1)(1-2\alpha) D_m - 2mP(\alpha^{m-1} - (1-\alpha)^{m-1}).$$

Fix $\alpha \in (0, \frac{1}{2})$, so $1-2\alpha > 0$ and $\alpha < 1-\alpha$; mP^{m-2}/D_m^3 is positive, so $\text{sign } s''_m = \text{sign } E_m$.

- If $m > 1$: the factor $m-1$ is positive, so the first term of E_m is positive; and since the exponent $m-1$ is positive, $\alpha^{m-1} < (1-\alpha)^{m-1}$, so the bracket is negative and the second term is positive too. Hence $E_m > 0$ and s_m is strictly convex.
- If $m < 1$: both inequalities reverse. The factor $m-1$ is negative; and because the exponent $m-1$ is now negative, then $\alpha^{m-1} > (1-\alpha)^{m-1}$ and the second term is negative as well. Hence $E_m < 0$ and s_m is strictly concave.
- If $m = 1$: $s_1(\alpha) = \alpha$ and $s''_1 \equiv 0$.

Step 3. The hypothesis $\kappa_B \geq 1 \geq \kappa_U$ is $m_B \leq 1 \leq m_U$, with at least one inequality strict. By Step 2, $s''_{m_U} \geq 0$ and $s''_{m_B} \leq 0$ on $(0, \frac{1}{2})$, and at least one is strict, so

$$h'' = s''_{m_U} - s''_{m_B} > 0 \quad \text{on } (0, \frac{1}{2}) :$$

h is strictly convex there, and strictly concave on $(\frac{1}{2}, 1)$ by Step 1.

Step 4. h is continuous on $[0, \frac{1}{2}]$, strictly convex on the interior, with $h(0) = h(\frac{1}{2}) = 0$ and $h < 0$ in between by (ii). A strictly convex function attains its minimum at a single point, and that point cannot be an endpoint since the endpoint values exceed the interior ones; hence there is a unique interior minimizer $\alpha^\dagger \in (0, \frac{1}{2})$. By Step 1, $1-\alpha^\dagger$ is the unique maximizer on $(\frac{1}{2}, 1)$. $\square$

F Proof of Proposition 6

Throughout, $x = \ln \frac{\alpha}{1-\alpha}$, $\bar{b} = \ln \frac{\beta}{1-\beta}$, $m = 1/\kappa$, and $\Lambda(z) = (1 + e^{-z})^{-1}$, so that by (18) a voter's diet is $\Lambda(mx + \bar{b})$ and

$$h(\alpha; \beta) = \Lambda(m_U x + \bar{b}) - \Lambda(m_B x + \bar{b}), \quad m_U = \frac{1}{\kappa_U} > m_B = \frac{1}{\kappa_B}.$$

Diets are evaluated in the large-electorate limit (Lemma 3). Note that $\alpha < \frac{1}{2}$ is equivalent to $x < 0$, and $\beta < \frac{1}{2}$ to $\bar{b} < 0$.

A general fold. Splitting the integral at the midpoint and substituting α with $1 - \alpha$ on the upper half,

$$\Delta(F) = \int_0^{1/2} \left[h(\alpha) f(\alpha) + h(1 - \alpha) f(1 - \alpha) \right] d\alpha. \quad (54)$$

Every result below is read off (54) under a different restriction. When $\beta = \frac{1}{2}$ the gap is antisymmetric (Lemma 3(iii)) and (54) collapses to the mirror form (23), in which only the population imbalance $f(\alpha) - f(1 - \alpha)$ stays. When instead F is symmetric, (54) collapses to

$$\Delta(F) = \int_0^{1/2} H(\alpha; \beta) f(\alpha) d\alpha, \quad H(\alpha; \beta) \equiv h(\alpha; \beta) + h(1 - \alpha; \beta), \quad (55)$$

Part (ii) is the statement that H has a constant sign, determined by the default alone.

Part (i): one-sided electorates

Suppose F is supported on $(0, \frac{1}{2}]$ and is not the point mass at $\frac{1}{2}$. By Lemma 3(ii), $h < 0$ on $(0, \frac{1}{2})$ for every β , and by Lemma 3(i), $h(\frac{1}{2}) = 0$. The part $h(\alpha)f(\alpha)$ is therefore non-positive throughout the support and strictly negative on $(0, \frac{1}{2})$, which carries positive mass because F is not degenerate at the endpoint. Hence $\Delta(F) = \mathbb{E}_F[h(\alpha)] < 0$. The case of support in $[\frac{1}{2}, 1)$ is identical with signs reversed.

The default is absent from this argument because Lemma 3(ii) holds for every β : unbundling always withdraws news from an entertainment-leaning voter, whatever mixture the medium supplies by default. What the default can do is change the balance between the two halves of the electorate, which is why it enters part (ii) and not this one.

Part (ii): balanced electorates and the default

Let F be symmetric around $\frac{1}{2}$, so (55) applies and it suffices to find a sign of $H(\alpha; \beta)$ on $(0, \frac{1}{2})$.

Step 1: H is the news consumed by a mirror-image pair. Fix $\alpha < \frac{1}{2}$ and consider the two voters with tastes α and $1 - \alpha$. Their diets are $\Lambda(mx + \bar{b})$ and $\Lambda(-mx + \bar{b})$ respectively, so their combined news consumption under technology m is

$$\varphi(m) \equiv \Lambda(mx + \bar{b}) + \Lambda(-mx + \bar{b}), \quad \text{and} \quad H(\alpha; \beta) = \varphi(m_U) - \varphi(m_B).$$

The question is therefore whether a mirror-image pair consumes more or less news in total when steering becomes cheaper.

Step 2: the two voters sit symmetrically around $\bar{b}$. Write $u = mx < 0$. The pair occupies the positions $\bar{b} + u$ and $\bar{b} - u$ on the log-odds line: symmetric about the default $\bar{b}$, and separated by $2|u|$, a distance that grows with m . Cheaper steering spreads the pair apart around the default; it does not move its centre.

Step 3: which voter moves more. Differentiating,

$$\frac{d\varphi}{du} = \Lambda'(\bar{b} + u) - \Lambda'(\bar{b} - u), \quad \Lambda'(z) = \Lambda(z)(1 - \Lambda(z)).$$

Evaluated at a voter's own position, Λ' is exactly the pass-through factor $t_n^*(1 - t_n^*)$ of (20): it is largest for a voter whose diet is close to balanced and vanishes towards the corners. Of the two members of the pair, therefore, the one with the more balanced diet responds more strongly to a change in technology. Since Λ is increasing with $\Lambda(0) = \frac{1}{2}$, a diet is closer to balanced precisely when its log-odds position is closer to the origin, and

$$(\bar{b} + u)^2 - (\bar{b} - u)^2 = 4\bar{b}u,$$

so $|\bar{b} + u| < |\bar{b} - u|$ if and only if $\bar{b}u < 0$. As $u < 0$ on $(0, \frac{1}{2})$, this holds precisely when $\bar{b} > 0$. Hence

$$\frac{d\varphi}{du} > 0 \iff \bar{b} > 0, \quad \frac{d\varphi}{du} = 0 \iff \bar{b} = 0.$$

Step 4: sign of H . Because $d\varphi/dm = x \times d\varphi/du$ with $x < 0$, the last display reverses:

$$\frac{d\varphi}{dm} < 0 \iff \bar{b} > 0.$$

Since $m_U > m_B$, this gives $\text{sign } H = -\text{sign } \bar{b} = \text{sign}(\frac{1}{2} - \beta)$, uniformly in $\alpha \in (0, \frac{1}{2})$. Substituting into (55) and using $f \geq 0$ delivers $\text{sign } \Delta(F) = \text{sign}(\frac{1}{2} - \beta)$, whatever the dispersion of F . When $\bar{b} = 0$ the derivative vanishes identically, φ is constant in m , and $\Delta(F) = 0$ — which is the antisymmetry of Lemma 3(iii) recovered as a special case.

Interpretation. With a news-heavy default, $\bar{b} > 0$, the entertainment-leaning member of the pair sits at $\bar{b} - |u|$, nearer the steep part of the logistic than his news-leaning counterpart at $\bar{b} + |u|$. Cheaper steering therefore moves him further than it moves her: he abandons more news than she takes up, and the pair's total falls. With an entertainment-heavy default the roles reverse. Only a symmetric default places the pair symmetrically about the point of maximum responsiveness, so that the two movements cancel exactly. Neutrality thus requires two symmetries — of tastes and of the default — and breaking either one alone is enough to move aggregate scrutiny.

Step 5: reflection. Lemma 3(iii) gives $h(1 - \alpha; 1 - \beta) = -h(\alpha; \beta)$. Integrating against f and substituting α with $1 - \alpha$,

$$\Delta(\check{F}; 1 - \beta) = -\Delta(F; \beta), \quad \check{f}(\alpha) \equiv f(1 - \alpha).$$

Reflecting tastes and the default together reverses the aggregate effect, so no generality is lost in restricting attention to $\beta \leq \frac{1}{2}$.

Part (iii): the threshold in the Beta family

Let $\alpha \sim \text{Beta}(\vartheta\mu, \vartheta(1 - \mu))$ and consider Δ as a function of the mean taste μ .

At both ends of the range the aggregate effect vanishes. As $\mu \rightarrow 0$ the distribution concentrates at $\alpha = 0$ and as $\mu \rightarrow 1$ at $\alpha = 1$; by Lemma 3(i), $h(0) = h(1) = 0$, so $\Delta \rightarrow 0$ in either

limit. Voters with no interest in news, and voters interested in nothing else, are unmoved by the technology, so an electorate composed entirely of one or the other registers no aggregate change.

Between those limits Δ is not zero. For small μ the mass lies predominantly below the midpoint, where $h < 0$, so $\Delta < 0$ by the argument of part (i); for large μ it lies above and $\Delta > 0$. It follows that $\mu \mapsto \Delta$ cannot be monotone: a monotone function vanishing at both endpoints would vanish identically. Instead Δ falls from zero to an interior minimum, rises through zero, reaches an interior maximum, and returns to zero. For instance, at $\beta = \frac{1}{2}$ and $\vartheta = 6$ the minimum is -0.198 near $\mu \approx 0.19$ and the maximum $+0.198$ near $\mu \approx 0.81$.

Uniqueness of the threshold therefore does not follow from monotonicity. What is required, is that Δ changes sign exactly once in the interior: there is a unique $\mu^*(\beta, \vartheta)$ with $\Delta < 0$ for $\mu < \mu^*$ and $\Delta > 0$ for $\mu > \mu^*$.

The limit $\mu^* \rightarrow \frac{1}{2}$ as $\vartheta \rightarrow \infty$ has an exact cause. As tastes concentrate, $\Delta(\mu, \vartheta, \beta) \rightarrow h(\mu; \beta)$, so the threshold converges to the interior zero of h itself, which by Lemma 3(i) is $\frac{1}{2}$ whatever the default.

The reason is that the default mix changes how large the technology gap is, but not where it changes sign. Technology gap is zero at 0, $\frac{1}{2}$ and 1 for every β , while the two lobes are unequal. For instance, at $\beta = 0.2$ the *escape* lobe reaches only -0.099 against a *specialisation* lobe of $+0.527$, and the asymmetry reverses at $\beta = 0.8$. An electorate concentrated at μ samples h at essentially a single point, so whether scrutiny rises or falls depends only on which side of the midpoint that point lies — a condition free of β . A dispersed electorate samples both lobes, whose relative weight does depend on the default mix, and the balance therefore changes at a different mean.

Part (iv): homogeneity deepens the loss

Let $\beta \leq \frac{1}{2}$, fix $\mu < \frac{1}{2}$, and suppose no voter leans towards news, so that tastes lie in $(0, \frac{1}{2}]$. On that interval h is strictly convex — proved at $\beta = \frac{1}{2}$ in Lemma 3(iv). Since $\Delta = \mathbb{E}_F[h(\alpha)]$, convexity yields two facts about the signed aggregate:

- (a) $\mathbb{E}_F[h(\alpha)] \geq h(\mu)$, strictly unless α is degenerate at μ (Jensen's inequality);
- (b) $\mathbb{E}_F[h(\alpha)]$ strictly increases under any mean-preserving spread of F .

It remains to restate (a) and (b) on magnitudes, as the proposition does. Both sides of (a) are negative: tastes lie in $(0, \frac{1}{2}]$, where $h < 0$ by Lemma 3(ii), so the average $\Delta = \mathbb{E}_F[h(\alpha)]$ is negative, and $h(\mu) < 0$ because $\mu < \frac{1}{2}$. Taking absolute values reverses an inequality between negative numbers — if $a \geq b$ with $a, b < 0$ then $|a| = -a \leq -b = |b|$ — so (a) becomes

$$|\Delta| \leq |h(\mu)|,$$

with equality only for a degenerate electorate. The same reversal turns (b) into the statement that any mean-preserving spread strictly reduces $|\Delta|$. Thus statement (a) implies $|\Delta| \leq |h(\mu)|$, with equality only for a degenerate electorate, and statement (b) implies: any mean-preserving spread strictly reduces $|\Delta|$.

The economics is the bend in h , not its depth: a voter sitting exactly at μ occupies the lowest point of a convex curve, so replacing her by two voters straddling μ moves both onto flatter ground and raises the average towards zero. Heterogeneity is a form of insurance against the technology, and homogeneity weakens it. $\square$

G Proof of Proposition 10

In this proof, G , Δ and S are evaluated in the large-electorate limit, where the civic term $\zeta = O(1/N)$ vanishes and the two environments differ only through κ . Recall the decomposition (30),

$$\Delta W = \underbrace{G(F)}_{\text{private gain}} + \underbrace{g(\tilde{t}_n^B + \Delta(F)) - g(\tilde{t}_n^B)}_{\text{public effect}}, \quad \tilde{t}_n^B = \tilde{t}_n(\kappa_B; F),$$

and that g is increasing and concave.

Part (i)

It is convenient to read κ as a price. Write

$$\mathcal{D}(t) = t \ln \frac{t}{\beta} + (1-t) \ln \frac{1-t}{1-\beta} \geq 0$$

for the *steering effort* a voter exerts in moving his diet from the default β to t — zero when $t = \beta$, and larger the further he pulls away. The voter's problem is then

$$\Pi(t, \kappa) = \underbrace{t \ln \alpha + (1-t) \ln(1-\alpha)}_{\text{content payoff}} - \underbrace{\kappa \mathcal{D}(t)}_{\text{steering bill}}, \quad v(\alpha; \kappa, \beta) = \max_{t \in [0,1]} \Pi(t, \kappa),$$

so κ is the price per unit of steering effort and $\mathcal{D}(t)$ is the quantity purchased. Unbundling is a fall in that price, and the private gain is the corresponding gain in consumer surplus.

Step 1: a unique optimum. Π is strictly concave in t , since $\partial^2 \Pi / \partial t^2 = -\kappa \left[\frac{1}{t} + \frac{1}{1-t} \right] < 0$, and $\partial \Pi / \partial t$ diverges to $+\infty$ as $t \rightarrow 0$ and to $-\infty$ as $t \rightarrow 1$. The optimal diet $t^*(\kappa)$ is therefore interior and unique, characterised by (18).

Step 2: how the price affects the voter. Raising κ affects v in two ways: it raises the bill on the steering already being done, and it induces the voter to steer less. Formally, Π is affine in κ with coefficient $-\mathcal{D}(t)$, so v is a maximum of affine functions of κ — hence convex — and, the maximiser being unique, differentiable, with

$$\frac{\partial v}{\partial \kappa} = -\mathcal{D}(t^*(\kappa)).$$

The voter loses, per unit increase in the price, exactly the amount of steering he is already doing.

Step 3: who is affected at all. $\mathcal{D}(t) \geq 0$ with equality only at $t = \beta$, so the price matters to a voter if and only if he departs from the default. By (18),

$$\ln \frac{t_n^*}{1-t_n^*} = \frac{1}{\kappa} \ln \frac{\alpha}{1-\alpha} + \ln \frac{\beta}{1-\beta},$$

so $t_n^* = \beta$ exactly when $\ln \frac{\alpha}{1-\alpha} = 0$, that is $\alpha = \frac{1}{2}$ — whatever the price and whatever the default. A voter with perfectly balanced tastes wants the default mix and buys no steering;

every other voter buys some, and pays for it. Hence $\partial v/\partial \kappa < 0$ for all $\alpha \neq \frac{1}{2}$, and $\partial v/\partial \kappa = 0$ at $\alpha = \frac{1}{2}$.

Step 4: the gain is the saving on the bill. Integrating Step 2 between the two prices,

$$\underbrace{v(\alpha; \kappa_U, \beta) - v(\alpha; \kappa_B, \beta)}_{\text{private gain}} = \int_{\kappa_U}^{\kappa_B} \mathcal{D}(t^*(\kappa)) d\kappa > 0 \quad \text{for } \alpha \neq \frac{1}{2}, \quad (56)$$

and the gain is zero at $\alpha = \frac{1}{2}$. The right-hand side is the area under the voter's demand curve for steering between the old and new prices: $\mathcal{D}(t^*(\kappa))$ is decreasing in κ . The private gain from unbundling is thus exactly the steering bill the voter no longer pays, and it is larger the further his taste lies from $\frac{1}{2}$, because such voters were steering more to begin with. Note that no restriction on β was used: the gain is positive under any default mix.

Step 5: symmetry. Let $\beta = \frac{1}{2}$. Then by (28), $v = \kappa \ln [\frac{1}{2}\alpha^{1/\kappa} + \frac{1}{2}(1-\alpha)^{1/\kappa}]$ is a symmetric function of the pair $(\alpha, 1-\alpha)$: replacing α by $1-\alpha$ merely interchanges its two arguments. Hence $v(1-\alpha; \kappa) = v(\alpha; \kappa)$ for every κ , and the gain inherits the symmetry. This is the only step that uses $\beta = \frac{1}{2}$; with an asymmetric default the two arguments carry unequal weights and symmetry fails.

Step 6: aggregation. By Steps 3–4 the gain is non-negative everywhere and strictly positive except at the single point $\alpha = \frac{1}{2}$. Hence $G(F) = \mathbb{E}_F[v(\alpha; \kappa_U) - v(\alpha; \kappa_B)] > 0$ for every taste distribution other than the point mass at $\frac{1}{2}$.

Part (ii)

Suppose first that F is symmetric around $\frac{1}{2}$ and $\beta = \frac{1}{2}$. Proposition 6(i) gives $\Delta(F) = 0$, so the public term in (30) vanishes identically and $\Delta W = G(F)$, which is strictly positive by part (i) unless F is degenerate at $\frac{1}{2}$ — a case excluded because it is not symmetric around $\frac{1}{2}$ in the relevant sense only if it is that point mass, for which $\Delta W = 0$. The Beta density is symmetric precisely when $\mu = \frac{1}{2}$, for every ϑ .

More generally, suppose $\Delta(F) \geq 0$. Since g is increasing, $g(\tilde{t}_n^B + \Delta) - g(\tilde{t}_n^B) \geq 0$, and adding $G(F) > 0$ from part (i) gives $\Delta W > 0$. By Proposition 6(ii), $\Delta(F) \geq 0$ holds in the Beta family for every $\mu \geq \frac{1}{2}$. Hence, the two channels reinforce rather than conflict, and no threshold arises.

The contrast with part (iii) is structural. The private gain integrates a function that is *symmetric* around $\frac{1}{2}$ (Step 5) and so survives aggregation over a balanced electorate; the scrutiny effect integrates the *antisymmetric* function h (Lemma 3(iii)) and vanishes there. It is this asymmetry between the two channels, not the size of either, that makes a balanced electorate an unambiguous winner.

Part (iii)

Rearranging (30), $\Delta W < 0$ if and only if

$$g(\tilde{t}_n^B) - g(\tilde{t}_n^B + \Delta(F)) > G(F), \quad (57)$$

which is the first statement. Both sides are well defined without any differentiability assumption, and by part (i) the right-hand side is strictly positive, so (57) can fail only if the public loss is strictly positive too — consistent with $\Delta(F) < 0$ and g increasing.

For the second statement, note first that $S(F)$ is well defined and finite: $G(F) > 0$ by part (i), and $|\Delta(F)| > 0$ by hypothesis. Since $g \in C^1$ and $\Delta(F) < 0$, the mean value theorem provides $\xi \in (\tilde{t}_n^B + \Delta(F), \tilde{t}_n^B)$ with

$$g(\tilde{t}_n^B) - g(\tilde{t}_n^B + \Delta(F)) = g'(\xi) |\Delta(F)|.$$

This step uses only the differentiability of g , not its functional form. Dividing by $|\Delta(F)| > 0$ turns (57) into $g'(\xi) > S(F)$. Both sides are rates per unit of aggregate news share — $S(F)$ in content utility, g' in public good. Finally, because g is concave, g' is non-increasing, and $\tilde{t}_n^B + \Delta(F) < \xi < \tilde{t}_n^B$ gives

$$g'(\tilde{t}_n^B) \leq g'(\xi) \leq g'(\tilde{t}_n^B + \Delta(F)),$$

so $g'(\tilde{t}_n^B) > S(F)$ is sufficient and $g'(\tilde{t}_n^B + \Delta(F)) > S(F)$ necessary. $\square$

H Pandering under general monitoring

Proposition 1. Fix an information environment summarized by aggregate news consumption t_n . Suppose that an incumbent of type $\theta \in \{g, b\}$ chooses effort $e \in [0, \bar{e}]$, and that the probability of re-election is $\rho_\theta(t_n, e)$. Assume:

(P1) (*More monitoring*) For each θ , ρ_θ is weakly increasing in t_n .

(P2) (*Effort improves re-election*) For each θ , ρ_θ is increasing in e and twice continuously differentiable in (t_n, e) .

(P3) (*Monitoring raises the return to effort, especially for bad types*) For all (t_n, e) ,

$$\frac{\partial^2 \rho_b(t_n, e)}{\partial t_n \partial e} \geq \frac{\partial^2 \rho_g(t_n, e)}{\partial t_n \partial e} \geq 0.$$

(P4) (*Well-behaved effort choice*) For each θ , the incumbent chooses effort to maximize

$$R \rho_\theta(t_n, e) - c_\theta(e),$$

where $R > 0$, $c_\theta(\cdot)$ is differentiable and weakly convex, and $\rho_\theta(t_n, e)$ is concave in e . Assume moreover that the objective is strictly concave in e whenever the optimum is interior.

Then the optimal effort $e_\theta^*(t_n)$ is weakly increasing in t_n for each θ , and the increase is weakly larger for the bad type than for the good type. In particular, for low levels of monitoring, equilibrium effort is interior and increasing in t_n : higher news consumption raises the marginal electoral return to effort, thereby strengthening the incumbent's incentives to exert effort (“pandering”). For sufficiently high t_n , effort reaches its upper bound $\bar{e}$ and becomes insensitive to further increases in monitoring.

Proof. Fix a type $\theta \in \{g, b\}$. The incumbent solves

$$\max_{e \in [0, \bar{e}]} R \rho_\theta(t_n, e) - c_\theta(e).$$

By (P4), the objective is concave in e , and strictly concave whenever the optimum is interior, so the best response is well defined and unique.

Consider first an interior optimum $e_\theta^*(t_n) \in (0, \bar{e})$. The first-order condition is

$$R \partial_e \rho_\theta(t_n, e_\theta^*) = c'_\theta(e_\theta^*).$$

Differentiating implicitly with respect to t_n yields

$$R \left(\partial_{te} \rho_\theta(t_n, e_\theta^*) + \partial_{ee} \rho_\theta(t_n, e_\theta^*) \frac{de_\theta^*}{dt_n} \right) = c''_\theta(e_\theta^*) \frac{de_\theta^*}{dt_n}.$$

Rearranging,

$$\frac{de_\theta^*}{dt_n} = \frac{R \partial_{te} \rho_\theta(t_n, e_\theta^*)}{c''_\theta(e_\theta^*) - R \partial_{ee} \rho_\theta(t_n, e_\theta^*)}.$$

Because c_θ is weakly convex, $c''_\theta(e_\theta^*) \geq 0$, and because ρ_θ is concave in e , $\partial_{ee} \rho_\theta(t_n, e_\theta^*) \leq 0$. Hence the denominator is weakly positive, and strictly positive under strict concavity of the objective. Therefore the sign of de_θ^*/dt_n is the sign of $\partial_{te} \rho_\theta(t_n, e_\theta^*)$.

By (P3), $\partial_{te} \rho_\theta(t_n, e) \geq 0$ for each type, so

$$\frac{de_\theta^*}{dt_n} \geq 0.$$

Moreover, since

$$\partial_{te} \rho_b(t_n, e) \geq \partial_{te} \rho_g(t_n, e),$$

the response of equilibrium effort to monitoring is weakly larger for the bad type than for the good type.

Finally, because effort is constrained to lie in $[0, \bar{e}]$, the unconstrained optimum may eventually exceed $\bar{e}$ as t_n rises. In that case the constraint binds and $e_\theta^*(t_n) = \bar{e}$, so further increases in t_n no longer affect effort. Thus, for low levels of monitoring, equilibrium effort is interior and increasing in t_n , whereas for sufficiently high t_n effort reaches its upper bound and becomes insensitive to further increases in monitoring. $\square$